Neural Computation -- 2014

96 article(s), all issues

1. Spike Train SIMilarity Space (SSIMS): A Framework for Single Neuron and Ensemble Data Analysis

Carlos E. Vargas-Irwin, David M. Brandman, Jonas B. Zimmermann, John P. Donoghue, Michael J. Black

https://doi.org/10.1162/neco_a_00684 | pp. 1-31

Increased emphasis on circuit level activity in the brain makes it necessary to have methods to visualize and evaluate large-scale ensemble activity beyond that revealed by raster-histograms or pairwise correlations. We present a method to evaluate the relative similarity of neural spiking patterns by combining spike train distance metrics with dimensionality reduction. Spike train distance metrics provide an estimate of similarity between activity patterns at multiple temporal resolutions. Vectors of pair-wise distances are used to represent the intrinsic relationships between multiple activity patterns at the level of single units or neuronal ensembles. Dimensionality reduction is then used to project the data into concise representations suitable for clustering analysis as well as exploratory visualization. Algorithm performance and robustness are evaluated using multielectrode ensemble activity data recorded in behaving primates. We demonstrate how spike train SIMilarity space (SSIMS) analysis captures the relationship between goal directions for an eight-directional reaching task and successfully segregates grasp types in a 3D grasping task in the absence of kinematic information. The algorithm enables exploration of virtually any type of neural spiking (time series) data, providing similarity-based clustering of neural activity states with minimal assumptions about potential information encoding models.

---

2. Optimizing the Representation of Orientation Preference Maps in Visual Cortex

Nicholas J. Hughes, Geoffrey J. Goodhill

https://doi.org/10.1162/neco_a_00687 | pp. 32-41

The colorful representation of orientation preference maps in primary visual cortex has become iconic. However, the standard representation is misleading because it uses a color mapping to indicate orientations based on the HSV (hue, saturation, value) color space, for which important perceptual features such as brightness, and not just hue, vary among orientations. This means that some orientations stand out more than others, conveying a distorted visual impression. This is particularly problematic for visualizing subtle biases caused by slight overrepresentation of some orientations due to, for example, stripe rearing. We show that displaying orientation maps with a color mapping based on a slightly modified version of the HCL (hue, chroma, lightness) color space, so that primarily only hue varies between orientations, leads to a more balanced visual impression. This makes it easier to perceive the true structure of this seminal example of functional brain architecture.

---

3. Topological Sparse Learning of Dynamic Form Patterns

Thomas Guthier, Volker Willert, Julian Eggert

https://doi.org/10.1162/neco_a_00670 | pp. 42-73

Motion is a crucial source of information for a variety of tasks in social interactions. The process of how humans recognize complex articulated movements such as gestures or face expressions remains largely unclear. There is an ongoing discussion if and how explicit low-level motion information, such as optical flow, is involved in the recognition process. Motivated by this discussion, we introduce a computational model that classifies the spatial configuration of gradient and optical flow patterns. The patterns are learned with an unsupervised learning algorithm based on translation-invariant nonnegative sparse coding called VNMF that extracts prototypical optical flow patterns shaped, for example, as moving heads or limb parts. A key element of the proposed system is a lateral inhibition term that suppresses activations of competing patterns in the learning process, leading to a low number of dominant and topological sparse activations. We analyze the classification performance of the gradient and optical flow patterns on three real-world human action recognition and one face expression recognition data set. The results indicate that the recognition of human actions can be achieved by gradient patterns alone, but adding optical flow patterns increases the classification performance. The combined patterns outperform other biological-inspired models and are competitive with current computer vision approaches.

---

4. Dynamics of Gamma Bursts in Local Field Potentials

Priscilla E. Greenwood, Mark D. McDonnell, Lawrence M. Ward

https://doi.org/10.1162/neco_a_00688 | pp. 74-103

In this letter, we provide a stochastic analysis of, and supporting simulation data for, a stochastic model of the generation of gamma bursts in local field potential (LFP) recordings by interacting populations of excitatory and inhibitory neurons. Our interest is in behavior near a fixed point of the stochastic dynamics of the model. We apply a recent limit theorem of stochastic dynamics to probe into details of this local behavior, obtaining several new results. We show that the stochastic model can be written in terms of a rotation multiplied by a two-dimensional standard Ornstein-Uhlenbeck (OU) process. Viewing the rewritten process in terms of phase and amplitude processes, we are able to proceed further in analysis. We demonstrate that gamma bursts arise in the model as excursions of the modulus of the OU process. The associated pair of stochastic phase and amplitude processes satisfies their own pair of stochastic differential equations, which indicates that large phase slips occur between gamma bursts. This behavior is mirrored in LFP data simulated from the original model. These results suggest that the rewritten model is a valid representation of the behavior near the fixed point for a wide class of models of oscillatory neural processes.

---

5. Spatiotemporal Conditional Inference and Hypothesis Tests for Neural Ensemble Spiking Precision

Matthew Tom Harrison, Asohan Amarasingham, Wilson Truccolo

https://doi.org/10.1162/neco_a_00681 | pp. 104-150

The collective dynamics of neural ensembles create complex spike patterns with many spatial and temporal scales. Understanding the statistical structure of these patterns can help resolve fundamental questions about neural computation and neural dynamics. Spatiotemporal conditional inference (STCI) is introduced here as a semiparametric statistical framework for investigating the nature of precise spiking patterns from collections of neurons that is robust to arbitrarily complex and nonstationary coarse spiking dynamics. The main idea is to focus statistical modeling and inference not on the full distribution of the data, but rather on

families of conditional distributions of precise spiking given different types of coarse spiking. The framework is then used to develop families of hypothesis tests for probing the spatiotemporal precision of spiking patterns. Relationships among different conditional distributions are used to improve multiple hypothesis-testing adjustments and design novel Monte Carlo spike resampling algorithms. Of special note are algorithms that can locally jitter spike times while still preserving the instantaneous peristimulus time histogram or the instantaneous total spike count from a group of recorded neurons. The framework can also be used to test whether first-order maximum entropy models with possibly random and time-varying parameters can account for observed patterns of spiking. STCI provides a detailed example of the generic principle of conditional inference, which may be applicable to other areas of neurostatistical analysis.

---

6. Toward a Multisubject Analysis of Neural Connectivity

Chris J. Oates, Lília Carolina Carneiro da Costa, Thomas E. Nichols

https://doi.org/10.1162/neco_a_00690 | pp. 151-170

Directed acyclic graphs (DAGs) and associated probability models are widely used to model neural connectivity and communication channels. In many experiments, data are collected from multiple subjects whose connectivities may differ but are likely to share many features. In such circumstances, it is natural to leverage similarity among subjects to improve statistical efficiency. The first exact algorithm for estimation of multiple related DAGs was recently proposed by Oates, Smith, Mukherjee, and Cussens (2014). In this letter we present examples and discuss implications of the methodology as applied to the analysis of fMRI data from a multisubject experiment. Elicitation of tuning parameters requires care, and we illustrate how this may proceed retrospectively based on technical replicate data. In addition to joint learning of subject-specific connectivity, we allow for heterogeneous collections of subjects and simultaneously estimate relationships between the subjects themselves. This letter aims to highlight the potential for exact estimation in the multisubject setting.

---

7. Using Multilayer Perceptron Computation to Discover Ideal Insect Olfactory Receptor Combinations in the Mosquito and Fruit Fly for an Efficient Electronic Nose

Luqman R. Bachtiar, Charles P. Unsworth, Richard D. Newcomb

https://doi.org/10.1162/neco_a_00691 | pp. 171-201

The model organism, Drosophila melanogaster, and the mosquito Anopheles gambiae use 60 and 79 odorant receptors, respectively, to sense their olfactory world. However, a commercial "electronic nose" in the form of an insect olfactory biosensor demands very low numbers of receptors at its front end of detection due to the difficulties of receptor/sensor integration and functionalization. In this letter, we demonstrate how computation via artificial neural networks (ANNs), in the form of multilayer perceptrons (MLPs), can be successfully incorporated as the signal processing back end of the biosensor to drastically reduce the number of receptors to three while still retaining 100% performance of odorant detection to that of a full complement of receptors. In addition, we provide a detailed performance comparison between D. melanogaster and A. gambiae odorant receptors and demonstrate that A. gambiae receptors provide superior olfaction detection performance over D. melanogaster for very low receptor numbers. The results from this study present the possibility of using the computation of MLPs to discover ideal biological olfactory

receptors for an olfactory biosensor device to provide maximum classification performance of unknown odorants.

---

8. Graph Degree Sequence Solely Determines the Expected Hopfield Network Pattern Stability

Daniel Berend, Shlomi Dolev, Ariel Hanemann

https://doi.org/10.1162/neco_a_00685 | pp. 202-210

We analyze the effect of network topology on the pattern stability of the Hopfield neural network in the case of general graphs. The patterns are randomly selected from a uniform distribution. We start the Hopfield procedure from some pattern v. An error in an entry e of v is the situation where, if the procedure is started at e, the value of e flips. Such an entry is an instability point. Note that we disregard the value at e by the end of the procedure, as well as what happens if we start the procedure from another pattern v' or another entry e' of v. We measure the instability of the system by the expected total number of instability points of all the patterns. Our main result is that the instability of the system does not depend on the exact topology of the underlying graph, but rather only on its degree sequence. Moreover, for a large number of nodes, the instability can be approximated by $m\Sigma ni=1(1 - \Phi(\sqrt{}\delta i/m–1))$, where is the standard normal distribution function and $\delta 1, \ldots, \delta n$ are the degrees of the nodes.

---

9. Efficient Training of Convolutional Deep Belief Networks in the Frequency Domain for Application to High-Resolution 2D and 3D Images

Tom Brosch, Roger Tam

https://doi.org/10.1162/neco_a_00682 | pp. 211-227

Deep learning has traditionally been computationally expensive, and advances in training methods have been the prerequisite for improving its efficiency in order to expand its application to a variety of image classification problems. In this letter, we address the problem of efficient training of convolutional deep belief networks by learning the weights in the frequency domain, which eliminates the time-consuming calculation of convolutions. An essential consideration in the design of the algorithm is to minimize the number of transformations to and from frequency space. We have evaluated the running time improvements using two standard benchmark data sets, showing a speed-up of up to 8 times on 2D images and up to 200 times on 3D volumes. Our training algorithm makes training of convolutional deep belief networks on 3D medical images with a resolution of up to 128×128×128 voxels practical, which opens new directions for using deep learning for medical image analysis.

---

10. Conditional Density Estimation with Dimensionality Reduction via Squared-Loss Conditional Entropy Minimization

Voot Tangkaratt, Ning Xie, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00683 | pp. 228-254

Regression aims at estimating the conditional mean of output given input. However, regression is not informative enough if the conditional density is multimodal, heteroskedastic, and asymmetric. In such a

case, estimating the conditional density itself is preferable, but conditional density estimation (CDE) is challenging in high-dimensional space. A naive approach to coping with high dimensionality is to first perform dimensionality reduction (DR) and then execute CDE. However, a two-step process does not perform well in practice because the error incurred in the first DR step can be magnified in the second CDE step. In this letter, we propose a novel single-shot procedure that performs CDE and DR simultaneously in an integrated way. Our key idea is to formulate DR as the problem of minimizing a squared-loss variant of conditional entropy, and this is solved using CDE. Thus, an additional CDE step is not needed after DR. We demonstrate the usefulness of the proposed method through extensive experiments on various data sets, including humanoid robot transition and computer art.

---

11. Subdiffusive Dynamics of Bump Attractors: Mechanisms and Functional Roles

Yang Qi, Michael Breakspear, Pulin Gong

https://doi.org/10.1162/neco_a_00698 | pp. 255-280

Bump attractors are localized activity patterns that can self-sustain after stimulus presentation, and they are regarded as the neural substrate for a host of perceptual and cognitive processes. One of the characteristic features of bump attractors is that they are neutrally stable, so that noisy inputs cause them to drift away from their initial locations, severely impairing the accuracy of bump location-dependent neural coding. Previous modeling studies of such noise-induced drifting activity of bump attractors have focused on normal diffusive dynamics, often with an assumption that noisy inputs are uncorrelated. Here we show that long-range temporal correlations and spatial correlations in neural inputs generated by multiple interacting bumps cause them to drift in an anomalous subdiffusive way. This mechanism for generating subdiffusive dynamics of bump attractors is further analyzed based on a generalized Langevin equation. We demonstrate that subdiffusive dynamics can significantly improve the coding accuracy of bump attractors, since the variance of the bump displacement increases sublinearly over time and is much smaller than that of normal diffusion. Furthermore, we reanalyze existing psychophysical data concerning the spread of recalled cue position in spatial working memory tasks and show that its variance increases sublinearly with time, consistent with subdiffusive dynamics of bump attractors. Based on the probability density function of bump position, we also show that the subdiffusive dynamics result in a long-tailed decay of firing rate, greatly extending the duration of persistent activity.

---

12. Reliability of Information-Based Integration of EEG and fMRI Data: A Simulation Study

Sara Assecondi, Dirk Ostwald, Andrew P. Bagshaw

https://doi.org/10.1162/neco_a_00695 | pp. 281-305

Most studies involving simultaneous electroencephalographic (EEG) and functional magnetic resonance imaging (fMRI) data rely on the first-order, affine-linear correlation of EEG and fMRI features within the framework of the general linear model. An alternative is the use of information-based measures such as mutual information and entropy, which can also detect higher-order correlations present in the data. The estimate of information-theoretic quantities might be influenced by several parameters, such as the numerosity of the sample, the amount of correlation between variables, and the discretization (or binning)

strategy of choice. While these issues have been investigated for invasive neurophysiological data and a number of bias-correction estimates have been developed, there has been no attempt to systematically examine the accuracy of information estimates for the multivariate distributions arising in the context of EEG-fMRI recordings. This is especially important given the differences between electrophysiological and EEG-fMRI recordings. In this study, we drew random samples from simulated bivariate and trivariate distributions, mimicking the statistical properties of EEG-fMRI data. We compared the estimated information shared by simulated random variables with its numerical value and found that the interaction between the binning strategy and the estimation method influences the accuracy of the estimate. Conditional on the simulation assumptions, we found that the equipopulated binning strategy yields the best and most consistent results across distributions and bias correction methods. We also found that within bias correction techniques, the asymptotically debiased (TPMC), the jackknife debiased (JD), and the best upper bound (BUB) approach give similar results, and those are consistent across distributions.

---

13. Active Inference, Evidence Accumulation, and the Urn Task

Thomas H. B. FitzGerald, Philipp Schwartenbeck, Michael Moutoussis, Raymond J. Dolan, Karl Friston
https://doi.org/10.1162/neco_a_00699 | pp. 306-328

Deciding how much evidence to accumulate before making a decision is a problem we and other animals often face, but one that is not completely understood. This issue is particularly important because a tendency to sample less information (often known as reflection impulsivity) is a feature in several psychopathologies, such as psychosis. A formal understanding of information sampling may therefore clarify the computational anatomy of psychopathology. In this theoretical letter, we consider evidence accumulation in terms of active (Bayesian) inference using a generic model of Markov decision processes. Here, agents are equipped with beliefs about their own behavior--in this case, that they will make informed decisions. Normative decision making is then modeled using variational Bayes to minimize surprise about choice outcomes. Under this scheme, different facets of belief updating map naturally onto the functional anatomy of the brain (at least at a heuristic level). Of particular interest is the key role played by the expected precision of beliefs about control, which we have previously suggested may be encoded by dopaminergic neurons in the midbrain. We show that manipulating expected precision strongly affects how much information an agent characteristically samples, and thus provides a possible link between impulsivity and dopaminergic dysfunction. Our study therefore represents a step toward understanding evidence accumulation in terms of neurobiologically plausible Bayesian inference and may cast light on why this process is disordered in psychopathology.

---

14. A Neural Mass Model with Direct and Indirect Excitatory Feedback Loops: Identification of Bifurcations and Temporal Dynamics

Aurélie Garnier, Alexandre Vidal, Clément Huneau, Habib Benali
https://doi.org/10.1162/neco_a_00696 | pp. 329-364

Neural mass modeling is a part of computational neuroscience that was developed to study the general behavior of a neuronal population. This type of mesoscopic model is able to generate output signals that are comparable to experimental data, such as electroencephalograms. Classically, neural mass models consider two interconnected populations: excitatory pyramidal cells and inhibitory interneurons. However,

many authors have included an excitatory feedback on the pyramidal cell population. Two distinct approaches have been developed: a direct feedback on the main pyramidal cell population and an indirect feedback via a secondary pyramidal cell population. In this letter, we propose a new neural mass model that couples these two approaches. We perform a detailed bifurcation analysis and present a glossary of dynamical behaviors and associated time series. Our study reveals that the model is able to generate particular realistic time series that were never pointed out in either simulated or experimental data. Finally, we aim to evaluate the effect of balance between both excitatory feedbacks on the dynamical behavior of the model. For this purpose, we compute the codimension 2 bifurcation diagrams of the system to establish a map of the repartition of dynamical behaviors in a direct versus indirect feedback parameter space. A perspective of this work is, from a given temporal series, to estimate the parameter value range, especially in terms of direct versus indirect excitatory feedback.

---

15. Mismatched Training and Test Distributions Can Outperform Matched Ones

Carlos Roberto González, Yaser S. Abu■Mostafa

https://doi.org/10.1162/neco_a_00697 | pp. 365-387

In learning theory, the training and test sets are assumed to be drawn from the same probability distribution. This assumption is also followed in practical situations, where matching the training and test distributions is considered desirable. Contrary to conventional wisdom, we show that mismatched training and test distributions in supervised learning can in fact outperform matched distributions in terms of the bottom line, the out-of-sample performance, independent of the target function in question. This surprising result has theoretical and algorithmic ramifications that we discuss.

---

16. Foundations of Support Constraint Machines

Giorgio Gnecco, Marco Gori, Stefano Melacci, Marcello Sanguineti

https://doi.org/10.1162/neco_a_00686 | pp. 388-480

The mathematical foundations of a new theory for the design of intelligent agents are presented. The proposed learning paradigm is centered around the concept of constraint, representing the interactions with the environment, and the parsimony principle. The classical regularization framework of kernel machines is naturally extended to the case in which the agents interact with a richer environment, where abstract granules of knowledge, compactly described by different linguistic formalisms, can be translated into the unified notion of constraint for defining the hypothesis set. Constrained variational calculus is exploited to derive general representation theorems that provide a description of the optimal body of the agent (i.e., the functional structure of the optimal solution to the learning problem), which is the basis for devising new learning algorithms. We show that regardless of the kind of constraints, the optimal body of the agent is a support constraint machine (SCM) based on representer theorems that extend classical results for kernel machines and provide new representations. In a sense, the expressiveness of constraints yields a semantic-based regularization theory, which strongly restricts the hypothesis set of classical regularization. Some guidelines to unify continuous and discrete computational mechanisms are given so as to accommodate in the same framework various kinds of stimuli, for example, supervised examples and logic predicates. The proposed view of learning from constraints incorporates classical learning from examples

and extends naturally to the case in which the examples are subsets of the input space, which is related to learning propositional logic clauses.

---

17. Natural Gradient Learning Algorithms for RBF Networks

Junsheng Zhao, Haikun Wei, Chi Zhang, Weiling Li, Weili Guo, Kanjian Zhang
https://doi.org/10.1162/neco_a_00689 | pp. 481-505

Radial basis function (RBF) networks are one of the most widely used models for function approximation and classification. There are many strange behaviors in the learning process of RBF networks, such as slow learning speed and the existence of the plateaus. The natural gradient learning method can overcome these disadvantages effectively. It can accelerate the dynamics of learning and avoid plateaus. In this letter, we assume that the probability density function (pdf) of the input and the activation function are gaussian. First, we introduce natural gradient learning to the RBF networks and give the explicit forms of the Fisher information matrix and its inverse. Second, since it is difficult to calculate the Fisher information matrix and its inverse when the numbers of the hidden units and the dimensions of the input are large, we introduce the adaptive method to the natural gradient learning algorithms. Finally, we give an explicit form of the adaptive natural gradient learning algorithm and compare it to the conventional gradient descent method. Simulations show that the proposed adaptive natural gradient method, which can avoid the plateaus effectively, has a good performance when RBF networks are used for nonlinear functions approximation.

---

18. Delay Differential Analysis of Electroencephalographic Data

Claudia Lainscsek, Manuel E. Hernandez, Howard Poizner, Terrence J. Sejnowski
https://doi.org/10.1162/neco_a_00656 | pp. 615-627

We propose a time-domain approach to detect frequencies, frequency couplings, and phases using nonlinear correlation functions. For frequency analysis, this approach is a multivariate extension of discrete Fourier transform, and for higher-order spectra, it is a linear and multivariate alternative to multidimensional fast Fourier transform of multidimensional correlations. This method can be applied to short and sparse time series and can be extended to cross-trial and cross-channel spectra (CTS) for electroencephalography data where multiple short data segments from multiple trials of the same experiment are available. There are two versions of CTS. The first one assumes some phase coherency across the trials, while the second one is independent of phase coherency. We demonstrate that the phase-dependent version is more consistent with event-related spectral perturbation analysis and traditional Morlet wavelet analysis. We show that CTS can be applied to short data windows and yields higher temporal resolution than traditional Morlet wavelet analysis. Furthermore, the CTS can be used to reconstruct the event-related potential using all linear components of the CTS.

---

19. Parametric Inference in the Large Data Limit Using Maximally Informative Models

Justin B. Kinney, Gurinder S. Atwal
https://doi.org/10.1162/neco_a_00568 | pp. 637-653

Motivated by data-rich experiments in transcriptional regulation and sensory neuroscience, we consider the following general problem in statistical inference: when exposed to a high-dimensional signal S, a system of interest computes a representation R of that signal, which is then observed through a noisy measurement M. From a large number of signals and measurements, we wish to infer the "filter" that maps S to R. However, the standard method for solving such problems, likelihood-based inference, requires perfect a priori knowledge of the "noise function" mapping R to M. In practice such noise functions are usually known only approximately, if at all, and using an incorrect noise function will typically bias the inferred filter. Here we show that in the large data limit, this need for a precharacterized noise function can be circumvented by searching for filters that instead maximize the mutual information I[M; R] between observed measurements and predicted representations. Moreover, if the correct filter lies within the space of filters being explored, maximizing mutual information becomes equivalent to simultaneously maximizing every dependence measure that satisfies the data processing inequality. It is important to note that maximizing mutual information will typically leave a small number of directions in parameter space unconstrained. We term these directions diffeomorphic modes and present an equation that allows these modes to be derived systematically. The presence of diffeomorphic modes reflects a fundamental and nontrivial substructure within parameter space, one that is obscured by standard likelihood-based inference.

---

20. Input Statistics and Hebbian Cross-Talk Effects

Anca R■dulescu

https://doi.org/10.1162/neco_a_00565 | pp. 654-692

As an extension of prior work, we studied inspecific Hebbian learning using the classical Oja model. We used a combination of analytical tools and numerical simulations to investigate how the effects of synaptic cross talk (which we also refer to as synaptic inspecificity) depend on the input statistics. We investigated a variety of patterns that appear in dimensions higher than two (and classified them based on covariance type and input bias). We found that the effects of cross talk on learning dynamics and outcome is highly dependent on the input statistics and that cross talk may lead in some cases to catastrophic effects on learning or development. Arbitrarily small levels of cross talk are able to trigger bifurcations in learning dynamics, or bring the system in close enough proximity to a critical state, to make the effects indistinguishable from a real bifurcation. We also investigated how cross talk behaves toward unbiased ("competitive") inputs and in which circumstances it can help the system productively resolve the competition. Finally, we discuss the idea that sophisticated neocortical learning requires accurate synaptic updates (similar to polynucleotide copying, which requires highly accurate replication). Since it is unlikely that the brain can completely eliminate cross talk, we support the proposal that is uses a neural mechanism that "proofreads" the accuracy of the updates, much as DNA proofreading lowers copying error rate.

---

21. Learning Nonlinear Statistical Regularities in Natural Images by Modeling the Outer Product of Image Intensities

Peng Qi, Xiaolin Hu

https://doi.org/10.1162/neco_a_00567 | pp. 693-711

It is well known that there exist nonlinear statistical regularities in natural images. Existing approaches for capturing such regularities always model the image intensities by assuming a parameterized distribution for

the intensities and learn the parameters. In the letter, we propose to model the outer product of image intensities by assuming a gaussian distribution for it. A two-layer structure is presented, where the first layer is nonlinear and the second layer is linear. Trained on natural images, the first-layer bases resemble the receptive fields of simple cells in the primary visual cortex (V1), while the second-layer units exhibit some properties of the complex cells in V1, including phase invariance and masking effect. The model can be seen as an approximation of the covariance model proposed in Karklin and Lewicki (2009) but has more robust and efficient learning algorithms.

---

22. Dissociable Forms of Repetition Priming: A Computational Model

Kirill Makukhin, Scott Bolland

https://doi.org/10.1162/neco_a_00569 | pp. 712-738

Nondeclarative memory and novelty processing in the brain is an actively studied field of neuroscience, and reducing neural activity with repetition of a stimulus (repetition suppression) is a commonly observed phenomenon. Recent findings of an opposite trend-specifically, rising activity for unfamiliar stimuli-question the generality of repetition suppression and stir debate over the underlying neural mechanisms. This letter introduces a theory and computational model that extend existing theories and suggests that both trends are, in principle, the rising and falling parts of an inverted U-shaped dependence of activity with respect to stimulus novelty that may naturally emerge in a neural network with Hebbian learning and lateral inhibition. We further demonstrate that the proposed model is sufficient for the simulation of dissociable forms of repetition priming using real-world stimuli. The results of our simulation also suggest that the novelty of stimuli used in neuroscientific research must be assessed in a particularly cautious way. The potential importance of the inverted-U in stimulus processing and its relationship to the acquisition of knowledge and competencies in humans is also discussed.

---

23. Refined Rademacher Chaos Complexity Bounds with Applications to the Multikernel Learning Problem

Yunwen Lei, Lixin Ding

https://doi.org/10.1162/neco_a_00566 | pp. 739-760

Estimating the Rademacher chaos complexity of order two is important for understanding the performance of multikernel learning (MKL) machines. In this letter, we develop a novel entropy integral for Rademacher chaos complexities. As compared to the previous bounds, our result is much improved in that it introduces an adjustable parameter ε to prohibit the divergence of the involved integral. With the use of the iteration technique in Steinwart and Scovel (2007), we also apply our Rademacher chaos complexity bound to the MKL problems and improve existing learning rates.

---

24. Large Margin Low Rank Tensor Analysis

Guoqiang Zhong, Mohamed Cheriet

https://doi.org/10.1162/neco_a_00570 | pp. 761-780

We present a supervised model for tensor dimensionality reduction, which is called large margin low rank tensor analysis (LMLRTA). In contrast to traditional vector representation-based dimensionality reduction methods, LMLRTA can take any order of tensors as input. And unlike previous tensor dimensionality reduction methods, which can learn only the low-dimensional embeddings with a priori specified dimensionality, LMLRTA can automatically and jointly learn the dimensionality and the low-dimensional representations from data. Moreover, LMLRTA delivers low rank projection matrices, while it encourages data of the same class to be close and of different classes to be separated by a large margin of distance in the low-dimensional tensor space. LMLRTA can be optimized using an iterative fixed-point continuation algorithm, which is guaranteed to converge to a local optimal solution of the optimization problem. We evaluate LMLRTA on an object recognition application, where the data are represented as 2D tensors, and a face recognition application, where the data are represented as 3D tensors. Experimental results show the superiority of LMLRTA over state-of-the-art approaches.

---

25. Large-Scale Linear RankSVM

Ching-pei Lee, Chih■Jen Lin

https://doi.org/10.1162/neco_a_00571 | pp. 781-817

Linear rankSVM is one of the widely used methods for learning to rank. Although its performance may be inferior to nonlinear methods such as kernel rankSVM and gradient boosting decision trees, linear rankSVM is useful to quickly produce a baseline model. Furthermore, following its recent development for classification, linear rankSVM may give competitive performance for large and sparse data. A great deal of works have studied linear rankSVM. The focus is on the computational efficiency when the number of preference pairs is large. In this letter, we systematically study existing works, discuss their advantages and disadvantages, and propose an efficient algorithm. We discuss different implementation issues and extensions with detailed experiments. Finally, we develop a robust linear rankSVM tool for public use.

---

26. A Transition to Sharp Timing in Stochastic Leaky Integrate-and-Fire Neurons Driven by Frozen Noisy Input

Thibaud Taillefumier, Marcelo O. Magnasco

https://doi.org/10.1162/neco_a_00577 | pp. 819-859

The firing activity of intracellularly stimulated neurons in cortical slices has been demonstrated to be profoundly affected by the temporal structure of the injected current (Mainen & Sejnowski, 1995). This suggests that the timing features of the neural response may be controlled as much by its own biophysical characteristics as by how a neuron is wired within a circuit. Modeling studies have shown that the interplay between internal noise and the fluctuations of the driving input controls the reliability and the precision of neuronal spiking (Cecchi et al., 2000 ; Tiesinga, 2002 ; Fellous, Rudolph, Destexhe, & Sejnowski, 2003). In order to investigate this interplay, we focus on the stochastic leaky integrate-and-fire neuron and identify the Hölder exponent H of the integrated input as the key mathematical property dictating the regime of firing of a single-unit neuron. We have recently provided numerical evidence (Taillefumier & Magnasco, 2013) for the existence of a phase transition when [Formula: see text] becomes less than the statistical Hölder exponent associated with internal gaussian white noise (H=1/2). Here we describe the theoretical and numerical framework devised for the study of a neuron that is periodically driven by frozen noisy inputs with exponent

H>0. In doing so, we account for the existence of a transition between two regimes of firing when H=1/2, and we show that spiking times have a continuous density when the Hölder exponent satisfies H>1/2. The transition at H=1/2 formally separates rate codes, for which the neural firing probability varies smoothly, from temporal codes, for which the neuron fires at sharply defined times regardless of the intensity of internal noise.

---

27. Sample Skewness as a Statistical Measurement of Neuronal Tuning Sharpness

Jason M. Samonds, Brian Potetz, Tai Sing Lee

https://doi.org/10.1162/neco_a_00582 | pp. 860-906

We propose using the statistical measurement of the sample skewness of the distribution of mean firing rates of a tuning curve to quantify sharpness of tuning. For some features, like binocular disparity, tuning curves are best described by relatively complex and sometimes diverse functions, making it difficult to quantify sharpness with a single function and parameter. Skewness provides a robust nonparametric measure of tuning curve sharpness that is invariant with respect to the mean and variance of the tuning curve and is straightforward to apply to a wide range of tuning, including simple orientation tuning curves and complex object tuning curves that often cannot even be described parametrically. Because skewness does not depend on a specific model or function of tuning, it is especially appealing to cases of sharpening where recurrent interactions among neurons produce sharper tuning curves that deviate in a complex manner from the feedforward function of tuning. Since tuning curves for all neurons are not typically well described by a single parametric function, this model independence additionally allows skewness to be applied to all recorded neurons, maximizing the statistical power of a set of data. We also compare skewness with other nonparametric measures of tuning curve sharpness and selectivity. Compared to these other nonparametric measures tested, skewness is best used for capturing the sharpness of multimodal tuning curves defined by narrow peaks (maximum) and broad valleys (minima). Finally, we provide a more formal definition of sharpness using a shape-based information gain measure and derive and show that skewness is correlated with this definition.

---

28. Prewhitening High-Dimensional fMRI Data Sets Without Eigendecomposition

Abd■Krim Seghouane, Yousef Saad

https://doi.org/10.1162/neco_a_00578 | pp. 907-919

This letter proposes an algorithm for linear whitening that minimizes the mean squared error between the original and whitened data without using the truncated eigendecomposition (ED) of the covariance matrix of the original data. This algorithm uses Lanczos vectors to accurately approximate the major eigenvectors and eigenvalues of the covariance matrix of the original data. The major advantage of the proposed whitening approach is its low computational cost when compared with that of the truncated ED. This gain comes without sacrificing accuracy, as illustrated with an experiment of whitening a high-dimensional fMRI data set.

---

29. Signal-Tuned Gabor Functions as Models for Stimulus-Dependent Cortical Receptive Fields

José R. A. Torreão, Silvia M. C. Victer, Marcos Serrou do Amaral

https://doi.org/10.1162/neco_a_00581 | pp. 920-952

We propose and analyze a model, based on signal-tuned Gabor functions, for the receptive fields and responses of V1 cells. Signal-tuned Gabor functions are gaussian-modulated sinusoids whose parameters are obtained from a given, spatial, or spectral "tuning" signal. These functions can be proven to yield exact representations of their tuning signals and have recently been proposed as the kernels of a variant Gabor transform-the signal-tuned Gabor transform (STGT)-which allows the accurate detection of spatial and spectral events. Here we show that by modeling the receptive fields of simple and complex cells as signal-tuned Gabor functions and expressing their responses as STGTs, we are able to replicate the properties of these cells when tested with standard grating and slit inputs, at the same time emulating their stimulus-dependent character as revealed by recent neurophysiological studies.

---

30. Energy Complexity of Recurrent Neural Networks

Ji■■■ Š■■ma

https://doi.org/10.1162/neco_a_00579 | pp. 953-973

Recently a new so-called energy complexity measure has been introduced and studied for feedforward perceptron networks. This measure is inspired by the fact that biological neurons require more energy to transmit a spike than not to fire, and the activity of neurons in the brain is quite sparse, with only about 1% of neurons firing. In this letter, we investigate the energy complexity of recurrent networks, which counts the number of active neurons at any time instant of a computation. We prove that any deterministic finite automaton with m states can be simulated by a neural network of optimal size [Formula: see text] with the time overhead of [Formula: see text] per one input bit, using the energy O(e), for any e such that [Formula: see text] and e=O(s), which shows the time-energy trade-off in recurrent networks. In addition, for the time overhead [Formula: see text] satisfying [Formula: see text], we obtain the lower bound of [Formula: see text] on the energy of such a simulation for some constant c>0 and for infinitely many s.

---

31. On Some Classes of Sequential Spiking Neural P Systems

Xingyi Zhang, Xiangxiang Zeng, Bin Luo, Linqiang Pan

https://doi.org/10.1162/neco_a_00580 | pp. 974-997

Spiking neural P systems (SN P systems) are a class of distributed parallel computing devices inspired by the way neurons communicate by means of spikes; neurons work in parallel in the sense that each neuron that can fire should fire, but the work in each neuron is sequential in the sense that at most one rule can be applied at each computation step. In this work, with biological inspiration, we consider SN P systems with the restriction that at each step, one of the neurons (i.e., sequential mode) or all neurons (i.e., pseudo-sequential mode) with the maximum (or minimum) number of spikes among the neurons that are active (can spike) will fire. If an active neuron has more than one enabled rule, it nondeterministically chooses one of the enabled rules to be applied, and the chosen rule is applied in an exhaustive manner (a kind of local parallelism): the rule is used as many times as possible. This strategy makes the system

sequential or pseudo-sequential from the global view of the whole network and locally parallel at the level of neurons. We obtain four types of SN P systems: maximum/minimum spike number induced sequential/pseudo-sequential SN P systems with exhaustive use of rules. We prove that SN P systems of these four types are all Turing universal as number-generating computation devices. These results illustrate that the restriction of sequentiality may have little effect on the computation power of SN P systems.

---

32. A Causal Perspective on the Analysis of Signal and Noise Correlations and Their Role in Population Coding

Daniel Chicharro

https://doi.org/10.1162/neco_a_00588 | pp. 999-1054

The role of correlations between neuronal responses is crucial to understanding the neural code. A framework used to study this role comprises a breakdown of the mutual information between stimuli and responses into terms that aim to account for different coding modalities and the distinction between different notions of independence. Here we complete the list of types of independence and distinguish activity independence (related to total correlations), conditional independence (related to noise correlations), signal independence (related to signal correlations), coding independence (related to information transmission), and information independence (related to redundancy). For each type, we identify the probabilistic criterion that defines it, indicate the information-theoretic measure used as statistic to test for it, and provide a graphical criterion to recognize the causal configurations of stimuli and responses that lead to its existence. Using this causal analysis, we first provide sufficiency conditions relating these types. Second, we differentiate the use of the measures as statistics to test for the existence of independence from their use for quantification. We indicate that signal and noise correlation cannot be quantified separately. Third, we explicitly define alternative system configurations used to construct the measures, in which noise correlations or noise and signal correlations are eliminated. Accordingly, we examine which measures are meaningful only as a comparison across configurations and which ones provide a characterization of the actually observed responses without resorting to other configurations. Fourth, we compare the commonly used nonparametric approach to eliminate noise correlations with a functional (model-based) approach, showing that the former approach does not remove those effects of noise correlations captured by the tuning properties of the individual neurons, and implies nonlocal causal structure manipulations. These results improve the interpretation of the measures on the framework and help in understanding how to apply it to analyze the role of correlations.

---

33. Toward Unified Hybrid Simulation Techniques for Spiking Neural Networks

Michiel D'Haene, Michiel Hermans, Benjamin Schrauwen

https://doi.org/10.1162/neco_a_00587 | pp. 1055-1079

In the field of neural network simulation techniques, the common conception is that spiking neural network simulators can be divided in two categories: time-step-based and event-driven methods. In this letter, we look at state-of-the art simulation techniques in both categories and show that a clear distinction between both methods is increasingly difficult to define. In an attempt to improve the weak points of each simulation method, ideas of the alternative method are, sometimes unknowingly, incorporated in the simulation engine.

Clearly the ideal simulation method is a mix of both methods. We formulate the key properties of such an efficient and generally applicable hybrid approach.

---

34. Neural Decoding with Kernel-Based Metric Learning

Austin J. Brockmeier, John Choi, Evan Kriminger, Joseph T. Francis, José C. Pr■■ncipe

https://doi.org/10.1162/neco_a_00591 | pp. 1080-1107

In studies of the nervous system, the choice of metric for the neural responses is a pivotal assumption. For instance, a well-suited distance metric enables us to gauge the similarity of neural responses to various stimuli and assess the variability of responses to a repeated stimulus-exploratory steps in understanding how the stimuli are encoded neurally. Here we introduce an approach where the metric is tuned for a particular neural decoding task. Neural spike train metrics have been used to quantify the information content carried by the timing of action potentials. While a number of metrics for individual neurons exist, a method to optimally combine single-neuron metrics into multineuron, or population-based, metrics is lacking. We pose the problem of optimizing multineuron metrics and other metrics using centered alignment, a kernel-based dependence measure. The approach is demonstrated on invasively recorded neural data consisting of both spike trains and local field potentials. The experimental paradigm consists of decoding the location of tactile stimulation on the forepaws of anesthetized rats. We show that the optimized metrics highlight the distinguishing dimensions of the neural response, significantly increase the decoding accuracy, and improve nonlinear dimensionality reduction methods for exploratory neural analysis.

---

35. Adaptive Multiclass Classification for Brain Computer Interfaces

Alberto Llera, Vicenç Gómez, Hilbert J. Kappen

https://doi.org/10.1162/neco_a_00592 | pp. 1108-1127

We consider the problem of multiclass adaptive classification for brain-computer interfaces and propose the use of multiclass pooled mean linear discriminant analysis (MPMLDA), a multiclass generalization of the adaptation rule introduced by Vidaurre, Kawanabe, von Bünau, Blankertz, and Müller (2010) for the binary class setting. Using publicly available EEG data sets and tangent space mapping (Barachant, Bonnet, Congedo, & Jutten, 2012) as a feature extractor, we demonstrate that MPMLDA can significantly outperform state-of-the-art multiclass static and adaptive methods. Furthermore, efficient learning rates can be achieved using data from different subjects.

---

36. On Nonnegative Matrix Factorization Algorithms for Signal-Dependent Noise with Application to Electromyography Data

Karthik Devarajan, Vincent C. K. Cheung

https://doi.org/10.1162/neco_a_00576 | pp. 1128-1168

Nonnegative matrix factorization (NMF) by the multiplicative updates algorithm is a powerful machine learning method for decomposing a high-dimensional nonnegative matrix V into two nonnegative matrices, W and H, where V ~ WH. It has been successfully applied in the analysis and interpretation of large-scale data arising in neuroscience, computational biology, and natural language processing, among other areas. A

distinctive feature of NMF is its nonnegativity constraints that allow only additive linear combinations of the data, thus enabling it to learn parts that have distinct physical representations in reality. In this letter, we describe an information-theoretic approach to NMF for signal-dependent noise based on the generalized inverse gaussian model. Specifically, we propose three novel algorithms in this setting, each based on multiplicative updates, and prove monotonicity of updates using the EM algorithm. In addition, we develop algorithm-specific measures to evaluate their goodness of fit on data. Our methods are demonstrated using experimental data from electromyography studies, as well as simulated data in the extraction of muscle synergies, and compared with existing algorithms for signal-dependent noise.

---

37. Direct Learning of Sparse Changes in Markov Networks by Density Ratio Estimation

Song Liu, John A. Quinn, Michael U. Gutmann, Taiji Suzuki, Masashi Sugiyama
https://doi.org/10.1162/neco_a_00589 | pp. 1169-1197

We propose a new method for detecting changes in Markov network structure between two sets of samples. Instead of naively fitting two Markov network models separately to the two data sets and figuring out their difference, we directly learn the network structure change by estimating the ratio of Markov network models. This density-ratio formulation naturally allows us to introduce sparsity in the network structure change, which highly contributes to enhancing interpretability. Furthermore, computation of the normalization term, a critical bottleneck of the naive approach, can be remarkably mitigated. We also give the dual formulation of the optimization problem, which further reduces the computation cost for large-scale Markov networks. Through experiments, we demonstrate the usefulness of our method.

---

38. Short-Term Memory Capacity in Networks via the Restricted Isometry Property

Adam S. Charles, Han Lun Yap, Christopher J. Rozell
https://doi.org/10.1162/neco_a_00590 | pp. 1198-1235

Cortical networks are hypothesized to rely on transient network activity to support short-term memory (STM). In this letter, we study the capacity of randomly connected recurrent linear networks for performing STM when the input signals are approximately sparse in some basis. We leverage results from compressed sensing to provide rigorous nonasymptotic recovery guarantees, quantifying the impact of the input sparsity level, the input sparsity basis, and the network characteristics on the system capacity. Our analysis demonstrates that network memory capacities can scale superlinearly with the number of nodes and in some situations can achieve STM capacities that are much larger than the network size. We provide perfect recovery guarantees for finite sequences and recovery bounds for infinite sequences. The latter analysis predicts that network STM systems may have an optimal recovery length that balances errors due to omission and recall mistakes. Furthermore, we show that the conditions yielding optimal STM capacity can be embodied in several network topologies, including networks with sparse or dense connectivities.

---

39. Errata to “Bayesian Community Detection” (Neural Computation, Sept. 2012, Vol. 24, No. 9:2434–2456)

Morten Mørup, Mikkel N. Schmidt

https://doi.org/10.1162/neco_c_00597 | pp. 1236-1237

Due to bugs in the MCMC sampler for the IHM, IDM, IRM, and IRMCB models, the inference and predictions for these models were not correct. We have therefore revised the sampler in the toolbox accompanying the article and rerun the experiments of Tables 1, 2, and 3.The new results show that the BCD outperforms the existing methods for about half of the considered networks, but not for the majority, as we originally stated (see the updated Table 2). In terms of the number of components extracted by IRM and BCD, it still holds in general that the BCD extracts more clusters; however, the new results show that it is not the case for all the considered networks (see the updated Table 1).

---

40. Motor Cortex Microcircuit Simulation Based on Brain Activity Mapping

George L. Chadderdon, Ashutosh Mohan, Benjamin A. Suter, Samuel A. Neymotin, Cliff C. Kerr, Joseph T. Francis, Gordon M. Shepherd, William W. Lytton

https://doi.org/10.1162/neco_a_00602 | pp. 1239-1262

The deceptively simple laminar structure of neocortex belies the complexity of intra- and interlaminar connectivity. We developed a computational model based primarily on a unified set of brain activity mapping studies of mouse M1. The simulation consisted of 775 spiking neurons of 10 cell types with detailed population-to-population connectivity. Static analysis of connectivity with graph-theoretic tools revealed that the corticostriatal population showed strong centrality, suggesting that would provide a network hub. Subsequent dynamical analysis confirmed this observation, in addition to revealing network dynamics that cannot be readily predicted through analysis of the wiring diagram alone. Activation thresholds depended on the stimulated layer. Low stimulation produced transient activation, while stronger activation produced sustained oscillations where the threshold for sustained responses varied by layer: 13% in layer 2/3, 54% in layer 5A, 25% in layer 5B, and 17% in layer 6. The frequency and phase of the resulting oscillation also depended on stimulation layer. By demonstrating the effectiveness of combined static and dynamic analysis, our results show how static brain maps can be related to the results of brain activity mapping.

---

41. Discovering Functional Neuronal Connectivity from Serial Patterns in Spike Train Data

Casey O. Diekman, Kohinoor Dasgupta, Vijay Nair, K. P. Unnikrishnan

https://doi.org/10.1162/neco_a_00598 | pp. 1263-1297

Repeating patterns of precisely timed activity across a group of neurons (called frequent episodes) are indicative of networks in the underlying neural tissue. This letter develops statistical methods to determine functional connectivity among neurons based on nonoverlapping occurrences of episodes. We study the distribution of episode counts and develop a two-phase strategy for identifying functional connections. For the first phase, we develop statistical procedures that are used to screen all two-node episodes and identify possible functional connections (edges). For the second phase, we develop additional statistical procedures to prune the two-node episodes and remove false edges that can be attributed to chains or fan-out

structures. The restriction to nonoverlapping occurrences makes the counting of all two-node episodes in phase 1 computationally efficient. The second (pruning) phase is critical since phase 1 can yield a large number of false connections. The scalability of the two-phase approach is examined through simulation. The method is then used to reconstruct the graph structure of observed neuronal networks, first from simulated data and then from recordings of cultured cortical neurons.

---

42. Risk-Sensitive Reinforcement Learning

Yun Shen, Michael J. Tobia, Tobias Sommer, Klaus Obermayer

https://doi.org/10.1162/neco_a_00600 | pp. 1298-1328

We derive a family of risk-sensitive reinforcement learning methods for agents, who face sequential decision-making tasks in uncertain environments. By applying a utility function to the temporal difference (TD) error, nonlinear transformations are effectively applied not only to the received rewards but also to the true transition probabilities of the underlying Markov decision process. When appropriate utility functions are chosen, the agents' behaviors express key features of human behavior as predicted by prospect theory (Kahneman & Tversky, 1979), for example, different risk preferences for gains and losses, as well as the shape of subjective probability curves. We derive a risk-sensitive Q-learning algorithm, which is necessary for modeling human behavior when transition probabilities are unknown, and prove its convergence. As a proof of principle for the applicability of the new framework, we apply it to quantify human behavior in a sequential investment task. We find that the risk-sensitive variant provides a significantly better fit to the behavioral data and that it leads to an interpretation of the subject's responses that is indeed consistent with prospect theory. The analysis of simultaneously measured fMRI signals shows a significant correlation of the risk-sensitive TD error with BOLD signal change in the ventral striatum. In addition we find a significant correlation of the risk-sensitive Q-values with neural activity in the striatum, cingulate cortex, and insula that is not present if standard Q-values are used.

---

43. On Criticality in High-Dimensional Data

Saeed Saremi, Terrence J. Sejnowski

https://doi.org/10.1162/neco_a_00607 | pp. 1329-1339

Data sets with high dimensionality such as natural images, speech, and text have been analyzed with methods from condensed matter physics. Here we compare recent approaches taken to relate the scale invariance of natural images to critical phenomena. We also examine the method of studying high-dimensional data through specific heat curves by applying the analysis to noncritical systems: 1D samples taken from natural images and 2D binary pink noise. Through these examples, we concluded that due to small sample sizes, specific heat is not a reliable measure for gauging whether high-dimensional data are critical. We argue that identifying order parameters and universality classes is a more reliable way to identify criticality in high-dimensional data.

---

44. Spiking Neural P Systems with Thresholds

Xiangxiang Zeng, Xingyi Zhang, Tao Song, Linqiang Pan

https://doi.org/10.1162/neco_a_00605 | pp. 1340-1361

Spiking neural P systems with weights are a new class of distributed and parallel computing models inspired by spiking neurons. In such models, a neuron fires when its potential equals a given value (called a threshold). In this work, spiking neural P systems with thresholds (SNPT systems) are introduced, where a neuron fires not only when its potential equals the threshold but also when its potential is higher than the threshold. Two types of SNPT systems are investigated. In the first one, we consider that the firing of a neuron consumes part of the potential (the amount of potential consumed depends on the rule to be applied). In the second one, once a neuron fires, its potential vanishes (i.e., it is reset to zero). The computation power of the two types of SNPT systems is investigated. We prove that the systems of the former type can compute all Turing computable sets of numbers and the systems of the latter type characterize the family of semilinear sets of numbers. The results show that the firing mechanism of neurons has a crucial influence on the computation power of the SNPT systems, which also answers an open problem formulated in Wang, Hoogeboom, Pan, P■un, and Pérez-Jiménez (2010).

---

45. Balanced Crossmodal Excitation and Inhibition Essential for Maximizing Multisensory Gain

Osamu Hoshino

https://doi.org/10.1162/neco_a_00606 | pp. 1362-1385

We examined whether and how the balancing of crossmodal excitation and inhibition affects intersensory facilitation. A neural network model, comprising lower-order unimodal networks (X, Y) and a higher-order multimodal network (M), was simulated. Crossmodal excitation was made by direct activation of principal cells of the X network by the Y network. Crossmodal inhibition was made in an indirect manner: the Y network activated glial cells of the X network. This let glial plasma membrane transporters export GABA molecules into the extracellular space and increased the level of ambient GABA. The ambient GABA molecules were accepted by extrasynaptic GABAa receptors and tonically inhibited principal cells of the X network. Namely, crossmodal inhibition was made through GABAergic gliotransmission. Intersensory facilitation was assessed in terms of multisensory gain: the difference between the numbers of spikes evoked by multisensory (XY) stimulation and unisensory (X-alone) stimulation. The maximal multisensory gain (XY-X) could be achieved at an intermediate noise level by balancing crossmodal excitation and inhibition. This result supports an experimentally derived conclusion: intersensory facilitation under noisy environmental conditions is not necessarily in accord with the principle of inverse effectiveness; rather, multisensory gain is maximal at intermediate signal-to-noise ratio (SNR) levels. The maximal multisensory gain was available at the weakest signal if noise was not present, indicating that the principle of inverse effectiveness is a special case of the intersensory facilitation model proposed here. We suggest that the balancing of crossmodal excitation and inhibition may be crucial for intersensory facilitation. The GABAergic glio-transmission-mediated crossmodal inhibitory mechanism effectively works for intersensory facilitation and on determining the maximal multisensory gain in the entire SNR range between the two extremes: low and high SNRs.

---

46. Universal Approximation Depth and Errors of Narrow Belief Networks with Discrete Units

Guido Montúfar

https://doi.org/10.1162/neco_a_00601 | pp. 1386-1407

We generalize recent theoretical work on the minimal number of layers of narrow deep belief networks that can approximate any probability distribution on the states of their visible units arbitrarily well. We relax the setting of binary units (Sutskever & Hinton, 2008 ; Le Roux & Bengio, 2008 , 2010 ; Montúfar & Ay, 2011) to units with arbitrary finite state spaces and the vanishing approximation error to an arbitrary approximation error tolerance. For example, we show that a q-ary deep belief network with L > or = 2 + (q[m-delta]-1 / (q-1)) layers of width n < or = + log(q) (m) + 1 for some [Formula : see text] can approximate any probability distribution on {0, 1, ... , q-1}n without exceeding a Kullback-Leibler divergence of delta. Our analysis covers discrete restricted Boltzmann machines and naive Bayes models as special cases.

---

47. Multiple Tests Based on a Gaussian Approximation of the Unitary Events Method with Delayed Coincidence Count

Christine Tuleau-Malot, Amel Rouis, Franck Grammont, Patricia Reynaud-Bouret

https://doi.org/10.1162/neco_a_00604 | pp. 1408-1454

The unitary events (UE) method is one of the most popular and efficient methods used over the past decade to detect patterns of coincident joint spike activity among simultaneously recorded neurons. The detection of coincidences is usually based on binned coincidence count (Grün, 1996), which is known to be subject to loss in synchrony detection (Grün, Diesmann, Grammont, Riehle, & Aertsen, 1999). This defect has been corrected by the multiple shift coincidence count (Grün et al., 1999). The statistical properties of this count have not been further investigated until this work, the formula being more difficult to deal with than the original binned count. First, we propose a new notion of coincidence count, the delayed coincidence count, which is equal to the multiple shift coincidence count when discretized point processes are involved as models for the spike trains. Moreover, it generalizes this notion to nondiscretized point processes, allowing us to propose a new gaussian approximation of the count. Since unknown parameters are involved in the approximation, we perform a plug-in step, where unknown parameters are replaced by estimated ones, leading to a modification of the approximating distribution. Finally the method takes the multiplicity of the tests into account via a Benjamini and Hochberg approach (Benjamini & Hochberg, 1995), to guarantee a prescribed control of the false discovery rate. We compare our new method, MTGAUE (multiple tests based on a gaussian approximation of the unitary events) and the UE method proposed in Grün et al. (1999) over various simulations, showing that MTGAUE extends the validity of the previous method. In particular, MTGAUE is able to detect both profusion and lack of coincidences with respect to the independence case and is robust to changes in the underlying model. Furthermore MTGAUE is applied on real data.

---

48. Intrinsic Graph Structure Estimation Using Graph Laplacian

Atsushi Noda, Hideitsu Hino, Masami Tatsuno, Shotaro Akaho, Noboru Murata

https://doi.org/10.1162/neco_a_00603 | pp. 1455-1483

A graph is a mathematical representation of a set of variables where some pairs of the variables are connected by edges. Common examples of graphs are railroads, the Internet, and neural networks. It is both theoretically and practically important to estimate the intensity of direct connections between variables. In

this study, a problem of estimating the intrinsic graph structure from observed data is considered. The observed data in this study are a matrix with elements representing dependency between nodes in the graph. The dependency represents more than direct connections because it includes influences of various paths. For example, each element of the observed matrix represents a co-occurrence of events at two nodes or a correlation of variables corresponding to two nodes. In this setting, spurious correlations make the estimation of direct connection difficult. To alleviate this difficulty, a digraph Laplacian is used for characterizing a graph. A generative model of this observed matrix is proposed, and a parameter estimation algorithm for the model is also introduced. The notable advantage of the proposed method is its ability to deal with directed graphs, while conventional graph structure estimation methods such as covariance selections are applicable only to undirected graphs. The algorithm is experimentally shown to be able to identify the intrinsic graph structure.

---

49. Causal Discovery via Reproducing Kernel Hilbert Space Embeddings

Zhitang Chen, Kun Zhang, Laiwan Chan, Bernhard Schölkopf

https://doi.org/10.1162/neco_a_00599 | pp. 1484-1517

Causal discovery via the asymmetry between the cause and the effect has proved to be a promising way to infer the causal direction from observations. The basic idea is to assume that the mechanism generating the cause distribution p(x) and that generating the conditional distribution p(y|x) correspond to two independent natural processes and thus p(x) and p(y|x) fulfill some sort of independence condition. However, in many situations, the independence condition does not hold for the anticausal direction; if we consider p(x, y) as generated via p(y)p(x|y), then there are usually some contrived mutual adjustments between p(y) and p(x|y). This kind of asymmetry can be exploited to identify the causal direction. Based on this postulate, in this letter, we define an uncorrelatedness criterion between p(x) and p(y|x) and, based on this uncorrelatedness, show asymmetry between the cause and the effect in terms that a certain complexity metric on p(x) and p(y|x) is less than the complexity metric on p(y) and p(x|y). We propose a Hilbert space embedding-based method EMD (an abbreviation for EMbeDding) to calculate the complexity metric and show that this method preserves the relative magnitude of the complexity metric. Based on the complexity metric, we propose an efficient kernel-based algorithm for causal discovery. The contribution of this letter is threefold. It allows a general transformation from the cause to the effect involving the noise effect and is applicable to both one-dimensional and high-dimensional data. Furthermore it can be used to infer the causal ordering for multiple variables. Extensive experiments on simulated and real-world data are conducted to show the effectiveness of the proposed method.

---

50. Bayesian Active Learning of Neural Firing Rate Maps with Transformed Gaussian Process Priors

Mijung Park, J. Patrick Weller, Gregory D. Horwitz, Jonathan W. Pillow

https://doi.org/10.1162/neco_a_00615 | pp. 1519-1541

A firing rate map, also known as a tuning curve, describes the nonlinear relationship between a neuron's spike rate and a low-dimensional stimulus (e.g., orientation, head direction, contrast, color). Here we investigate Bayesian active learning methods for estimating firing rate maps in closed-loop neurophysiology experiments. These methods can accelerate the characterization of such maps through the intelligent,

adaptive selection of stimuli. Specifically, we explore the manner in which the prior and utility function used in Bayesian active learning affect stimulus selection and performance. Our approach relies on a flexible model that involves a nonlinearly transformed gaussian process (GP) prior over maps and conditionally Poisson spiking. We show that infomax learning, which selects stimuli to maximize the information gain about the firing rate map, exhibits strong dependence on the seemingly innocuous choice of nonlinear transformation function. We derive an alternate utility function that selects stimuli to minimize the average posterior variance of the firing rate map and analyze the surprising relationship between prior parameterization, stimulus selection, and active learning performance in GP-Poisson models. We apply these methods to color tuning measurements of neurons in macaque primary visual cortex.

---

51. How to Compress Sequential Memory Patterns into Periodic Oscillations: General Reduction Rules

Kechen Zhang

https://doi.org/10.1162/neco_a_00618 | pp. 1542-1599

A neural network with symmetric reciprocal connections always admits a Lyapunov function, whose minima correspond to the memory states stored in the network. Networks with suitable asymmetric connections can store and retrieve a sequence of memory patterns, but the dynamics of these networks cannot be characterized as readily as that of the symmetric networks due to the lack of established general methods. Here, a reduction method is developed for a class of asymmetric attractor networks that store sequences of activity patterns as associative memories, as in a Hopfield network. The method projects the original activity pattern of the network to a low-dimensional space such that sequential memory retrievals in the original network correspond to periodic oscillations in the reduced system. The reduced system is self-contained and provides quantitative information about the stability and speed of sequential memory retrieval in the original network. The time evolution of the overlaps between the network state and the stored memory patterns can also be determined from extended reduced systems. The reduction procedure can be summarized by a few reduction rules, which are applied to several network models, including coupled networks and networks with time-delayed connections, and the analytical solutions of the reduced systems are confirmed by numerical simulations of the original networks. Finally, a local learning rule that provides an approximation to the connection weights involving the pseudoinverse is also presented.

---

52. The Competing Benefits of Noise and Heterogeneity in Neural Coding

Eric Hunsberger, Matthew P. Scott, Chris Eliasmith

https://doi.org/10.1162/neco_a_00621 | pp. 1600-1623

Noise and heterogeneity are both known to benefit neural coding. Stochastic resonance describes how noise, in the form of random fluctuations in a neuron's membrane voltage, can improve neural representations of an input signal. Neuronal heterogeneity refers to variation in any one of a number of neuron parameters and is also known to increase the information content of a population. We explore the interaction between noise and heterogeneity and find that their benefits to neural coding are not independent. Specifically, a neuronal population better represents an input signal when either noise or heterogeneity is added, but adding both does not always improve representation further. To explain this phenomenon, we propose that noise and heterogeneity operate using two shared mechanisms: (1)

temporally desynchronizing the firing of neurons in the population and (2) linearizing the response of a population to a stimulus. We first characterize the effects of noise and heterogeneity on the information content of populations of either leaky integrate-and-fire or FitzHugh-Nagumo neurons. We then examine how the mechanisms of desynchronization and linearization produce these effects, and find that they work to distribute information equally across all neurons in the population in terms of both signal timing (desynchronization) and signal amplitude (linearization). Without noise or heterogeneity, all neurons encode the same aspects of the input signal; adding noise or heterogeneity allows neurons to encode complementary aspects of the input signal, thereby increasing information content. The simulations detailed in this letter highlight the importance of heterogeneity and noise in population coding, demonstrate their complex interactions in terms of the information content of neurons, and explain these effects in terms of underlying mechanisms.

---

53. Anatomical Constraints on Lateral Competition in Columnar Cortical Architectures

Dylan R. Muir, Matthew Cook

https://doi.org/10.1162/neco_a_00613 | pp. 1624-1666

Competition is a well-studied and powerful mechanism for information processing in neuronal networks, providing noise rejection, signal restoration, decision making and associative memory properties, with relatively simple requirements for network architecture. Models based on competitive interactions have been used to describe the shaping of functional properties in visual cortex, as well as the development of functional maps in columnar cortex. These models require competition within a cortical area to occur on a wider spatial scale than cooperation, usually implemented by lateral inhibitory connections having a longer range than local excitatory connections. However, measurements of cortical anatomy reveal that the spatial extent of inhibition is in fact more restricted than that of excitation. Relatively few models reflect this, and it is unknown whether lateral competition can occur in cortical-like networks that have a realistic spatial relationship between excitation and inhibition. Here we analyze simple models for cortical columns and perform simulations of larger models to show how the spatial scales of excitation and inhibition can interact to produce competition through disynaptic inhibition. Our findings give strong support to the direct coupling effect-that the presence of competition across the cortical surface is predicted well by the anatomy of direct excitatory and inhibitory coupling and that multisynaptic network effects are negligible. This implies that for networks with short-range inhibition and longer-range excitation, the spatial extent of competition is even narrower than the range of inhibitory connections. Our results suggest the presence of network mechanisms that focus on intra-rather than intercolumn competition in neocortex, highlighting the need for both new models and direct experimental characterizations of lateral inhibition and competition in columnar cortex.

---

54. ROC-Based Estimates of Neural-Behavioral Covariations Using Matched Filters

Kamal Farah, Jackson E. T. Smith, Erik P. Cook

https://doi.org/10.1162/neco_a_00616 | pp. 1667-1689

Correlations between responses in visual cortex and perceptual performance help draw a functional link between neural activity and visually guided behavior. These correlations are commonly derived with

ROC-based neural-behavioral covariances (referred to as choice or detect probability) using boxcar analysis windows. Although boxcar windows capture the covariation between neural activity and behavior during steady-state stimulus presentations, they are not optimized to capture these correlations during short time-varying visual inputs. In this study, we implemented a matched-filter technique, combined with cross-validation, to improve the estimation of ROC-based neural-behavioral covariance under short and dynamic stimulus conditions. We show that this approach maximizes the area under the ROC curve and converges to the true neural-behavioral covariance using a Poisson spiking model. We also demonstrate that the matched filter, combined with cross-validation, reveals the dynamics of the neural-behavioral covariations of individual MT neurons during the detection of a brief motion stimulus.

---

55. Tonically Balancing Intracortical Excitation and Inhibition by GABAergic Gliotransmission

Meihong Zheng, Takami Matsuo, Ai Miyamoto, Osamu Hoshino

https://doi.org/10.1162/neco_a_00612 | pp. 1690-1716

For sensory cortices to respond reliably to feature stimuli, the balancing of neuronal excitation and inhibition is crucial. A typical example might be the balancing of phasic excitation within cell assemblies and phasic inhibition between cell assemblies. The former controls the gain of and the latter the tuning of neuronal responses. A change in ambient GABA concentration might affect the dynamic behavior of neurons in a tonic manner. For instance, an increase in ambient GABA concentration enhances the activation of extrasynaptic receptors, augments an inhibitory current, and thus inhibits neurons. When a decrease in ambient GABA concentration occurs, the tonic inhibitory current is reduced, and thus the neurons are relatively excited. We simulated a neural network model in order to examine whether and how such a tonic excitatory-inhibitory mechanism could work for sensory information processing. The network consists of cell assemblies. Each cell assembly, comprising principal cells (P), GABAergic interneurons (Ia, Ib), and glial cells (glia), responds to one particular feature stimulus. GABA transporters, embedded in glial plasma membranes, regulate ambient GABA levels. Hypothetical neuron-glia signaling via inhibitory (Ia-to-glia) and excitatory (P-to-glia) synaptic contacts was assumed. The former let transporters import (remove) GABA from the extracellular space and excited stimulus-relevant P cells. The latter let them export GABA into the extracellular space and inhibited stimulus-irrelevant P cells. The main finding was that the glial membrane transporter gave a combinatorial excitatory-inhibitory effect on P cells in a tonic manner, thereby improving the gain and tuning of neuronal responses. Interestingly, it worked cooperatively with the conventional, phasic excitatory-inhibitory mechanism. We suggest that the GABAergic gliotransmission mechanism may provide balanced intracortical excitation and inhibition so that the best perceptual performance of the cortex can be achieved.

---

56. Information-Theoretic Semi-Supervised Metric Learning via Entropy Regularization

Gang Niu, Bo Dai, Makoto Yamada, Masashi Sugiyama

https://doi.org/10.1162/neco_a_00614 | pp. 1717-1762

We propose a general information-theoretic approach to semi-supervised metric learning called SERAPH (SEmi-supervised metRic leArning Paradigm with Hypersparsity) that does not rely on the manifold

assumption. Given the probability parameterized by a Mahalanobis distance, we maximize its entropy on labeled data and minimize its entropy on unlabeled data following entropy regularization. For metric learning, entropy regularization improves manifold regularization by considering the dissimilarity information of unlabeled data in the unsupervised part, and hence it allows the supervised and unsupervised parts to be integrated in a natural and meaningful way. Moreover, we regularize SERAPH by trace-norm regularization to encourage low-dimensional projections associated with the distance metric. The nonconvex optimization problem of SERAPH could be solved efficiently and stably by either a gradient projection algorithm or an EM-like iterative algorithm whose M-step is convex. Experiments demonstrate that SERAPH compares favorably with many well-known metric learning methods, and the learned Mahalanobis distance possesses high discriminability even under noisy environments.

---

57. Incremental Learning by Message Passing in Hierarchical Temporal Memory

Erik M. Rehn, Davide Maltoni

https://doi.org/10.1162/neco_a_00617 | pp. 1763-1809

Hierarchical temporal memory (HTM) is a biologically inspired framework that can be used to learn invariant representations of patterns in a wide range of applications. Classical HTM learning is mainly unsupervised, and once training is completed, the network structure is frozen, thus making further training (i.e., incremental learning) quite critical. In this letter, we develop a novel technique for HTM (incremental) supervised learning based on gradient descent error minimization. We prove that error backpropagation can be naturally and elegantly implemented through native HTM message passing based on belief propagation. Our experimental results demonstrate that a two-stage training approach composed of unsupervised pretraining and supervised refinement is very effective (both accurate and efficient). This is in line with recent findings on other deep architectures.

---

58. Continuous Closed-Loop Decoder Adaptation with a Recursive Maximum Likelihood Algorithm Allows for Rapid Performance Acquisition in Brain-Machine Interfaces

Siddharth Dangi, Suraj Gowda, Helene G. Moorman, Amy L. Orsborn, Kelvin So, Maryam M. Shanechi, Jose M. Carmena

https://doi.org/10.1162/neco_a_00632 | pp. 1811-1839

Closed-loop decoder adaptation (CLDA) is an emerging paradigm for both improving and maintaining online performance in brain-machine interfaces (BMIs). The time required for initial decoder training and any subsequent decoder recalibrations could be potentially reduced by performing continuous adaptation, in which decoder parameters are updated at every time step during these procedures, rather than waiting to update the decoder at periodic intervals in a more batch-based process. Here, we present recursive maximum likelihood (RML), a CLDA algorithm that performs continuous adaptation of a Kalman filter decoder's parameters. We demonstrate that RML possesses a variety of useful properties and practical algorithmic advantages. First, we show how RML leverages the accuracy of updates based on a batch of data while still adapting parameters on every time step. Second, we illustrate how the RML algorithm is parameterized by a single, intuitive half-life parameter that can be used to adjust the rate of adaptation in

real time. Third, we show how even when the number of neural features is very large, RML's memory-efficient recursive update rules can be reformulated to also be computationally fast so that continuous adaptation is still feasible. To test the algorithm in closed-loop experiments, we trained three macaque monkeys to perform a center-out reaching task by using either spiking activity or local field potentials to control a 2D computer cursor. RML achieved higher levels of performance more rapidly in comparison to a previous CLDA algorithm that adapts parameters on a more intermediate timescale. Overall, our results indicate that RML is an effective CLDA algorithm for achieving rapid performance acquisition using continuous adaptation.

---

59. Neuronal Spike Train Entropy Estimation by History Clustering

Nicholas Watters, George N. Reeke

https://doi.org/10.1162/neco_a_00627 | pp. 1840-1872

Neurons send signals to each other by means of sequences of action potentials (spikes). Ignoring variations in spike amplitude and shape that are probably not meaningful to a receiving cell, the information content, or entropy of the signal depends on only the timing of action potentials, and because there is no external clock, only the interspike intervals, and not the absolute spike times, are significant. Estimating spike train entropy is a difficult task, particularly with small data sets, and many methods of entropy estimation have been proposed. Here we present two related model-based methods for estimating the entropy of neural signals and compare them to existing methods. One of the methods is fast and reasonably accurate, and it converges well with short spike time records; the other is impractically time-consuming but apparently very accurate, relying on generating artificial data that are a statistical match to the experimental data. Using the slow, accurate method to generate a best-estimate entropy value, we find that the faster estimator converges to this value more closely and with smaller data sets than many existing entropy estimators.

---

60. Memory Nearly on a Spring: A Mean First Passage Time Approach to Memory Lifetimes

Terry Elliott

https://doi.org/10.1162/neco_a_00622 | pp. 1873-1923

We study memory lifetimes in a perceptron-based framework with binary synapses, using the mean first passage time for the perceptron's total input to fall below firing threshold to define memory lifetimes. Working with the simplest memory-related model of synaptic plasticity, we may obtain exact results for memory lifetimes or, working in the continuum limit, good analytical approximations that afford either much qualitative insight or extremely good quantitative agreement. In one particular limit, we find that memory dynamics reduce to the well-understood Ornstein-Uhlenbeck process. We show that asymptotically, the lifetimes of memories grow logarithmically in the number of synapses when the perceptron's firing threshold is zero, reproducing standard results from signal-to-noise ratio analyses. However, this is only an asymptotically valid result, and we show that extending its application outside the range of its validity leads to a massive overestimate of the minimum number of synapses required for successful memory encoding. In the case that the perceptron's firing threshold is positive, we find the remarkable result that memory lifetimes are strictly bounded from above. Asymptotically, the dependence of memory lifetimes on the number of synapses drops out entirely, and this asymptotic result provides a strict upper bound on memory lifetimes

away from this asymptotic regime. The classic logarithmic growth of memory lifetimes in the simplest, palimpsest memories is therefore untypical and nongeneric: memory lifetimes are typically strictly bounded from above.

---

61. Sparseness, Antisparseness and Anything in Between: The Operating Point of a Neuron Determines Its Computational Repertoire

Terry Elliott

https://doi.org/10.1162/neco_a_00630 | pp. 1924-1972

A recent model of intrinsic plasticity coupled to Hebbian synaptic plasticity proposes that adaptation of a neuron's threshold and gain in a sigmoidal response function to achieve a sparse, exponential output firing rate distribution facilitates the discovery of heavy-tailed or super- gaussian sources in the neuron's inputs. We show that the exponential output distribution is irrelevant to these dynamics and that, furthermore, while sparseness is sufficient, it is not necessary. The intrinsic plasticity mechanism drives the neuron's threshold large and positive, and we prove that in such a regime, the neuron will find supergaussian sources; equally, however, if the threshold is large and negative (an antisparse regime), it will also find supergaussian sources. Away from such extremes, the neuron can also discover subgaussian sources. By examining a neuron with a fixed sigmoidal nonlinearity and considering the synaptic strength fixed-point structure in the two-dimensional parameter space defined by the neuron's threshold and gain, we show that this space is carved up into sub- and supergaussian-input-finding regimes, possibly with regimes of simultaneous stability of sub- and supergaussian sources or regimes of instability of all sources; a single gaussian source may also be stabilized by the presence of a nongaussian source. A neuron's operating point (essentially its threshold and gain coupled with its input statistics) therefore critically determines its computational repertoire. Intrinsic plasticity mechanisms induce trajectories in this parameter space but do not fundamentally modify it. Unless the trajectories cross critical boundaries in this space, intrinsic plasticity is irrelevant and the neuron's nonlinearity may be frozen with identical receptive field refinement dynamics.

---

62. Sequential Activity in Asymmetrically Coupled Winner-Take-All Circuits

Hesham Mostafa, Giacomo Indiveri

https://doi.org/10.1162/neco_a_00619 | pp. 1973-2004

Understanding the sequence generation and learning mechanisms used by recurrent neural networks in the nervous system is an important problem that has been studied extensively. However, most of the models proposed in the literature are either not compatible with neuroanatomy and neurophysiology experimental findings, or are not robust to noise and rely on fine tuning of the parameters. In this work, we propose a novel model of sequence learning and generation that is based on the interactions among multiple asymmetrically coupled winner-take-all (WTA) circuits. The network architecture is consistent with mammalian cortical connectivity data and uses realistic neuronal and synaptic dynamics that give rise to noise-robust patterns of sequential activity. The novel aspect of the network we propose lies in its ability to produce robust patterns of sequential activity that can be halted, resumed, and readily modulated by external input, and in its ability to make use of realistic plastic synapses to learn and reproduce the arbitrary input-imposed sequential patterns. Sequential activity takes the form of a single activity bump that stably propagates through multiple WTA circuits along one of a number of possible paths. Because the network

can be configured to either generate spontaneous sequences or wait for external inputs to trigger a transition in the sequence, it provides the basis for creating state-dependent perception-action loops. We first analyze a rate-based approximation of the proposed spiking network to highlight the relevant features of the network dynamics and then show numerical simulation results with spiking neurons, realistic conductance-based synapses, and spike-timing dependent plasticity (STDP) rules to validate the rate-based model.

---

63. Synchronization of Stochastic Competitive Neural Networks with Different Timescales and Reaction-Diffusion Terms

Yanchao Shi, Peiyong Zhu

https://doi.org/10.1162/neco_a_00629 | pp. 2005-2024

We propose a feedback controller for the synchronization of stochastic competitive neural networks with different timescales and reaction-diffusion terms. By constructing a proper Lyapunov-Krasovskii functional, as well as employing stochastic analysis theory, the LaShall-type invariance principle for stochastic differential delay equations, and a linear matrix inequality (LMI) technique, a feedback controller is designed to achieve the asymptotical synchronization of coupled stochastic competitive neural networks. A simulation example is given to show the effectiveness of the theoretical results.

---

64. A Semiparametric Bayesian Model for Detecting Synchrony Among Multiple Neurons

Babak Shahbaba, Bo Zhou, Shiwei Lan, Hernando Ombao, David Moorman, Sam Behseta

https://doi.org/10.1162/neco_a_00631 | pp. 2025-2051

We propose a scalable semiparametric Bayesian model to capture dependencies among multiple neurons by detecting their cofiring (possibly with some lag time) patterns over time. After discretizing time so there is at most one spike at each interval, the resulting sequence of 1s (spike) and 0s (silence) for each neuron is modeled using the logistic function of a continuous latent variable with a gaussian process prior. For multiple neurons, the corresponding marginal distributions are coupled to their joint probability distribution using a parametric copula model. The advantages of our approach are as follows. The nonparametric component (i.e., the gaussian process model) provides a flexible framework for modeling the underlying firing rates, and the parametric component (i.e., the copula model) allows us to make inferences regarding both contemporaneous and lagged relationships among neurons. Using the copula model, we construct multivariate probabilistic models by separating the modeling of univariate marginal distributions from the modeling of a dependence structure among variables. Our method is easy to implement using a computationally efficient sampling algorithm that can be easily extended to high-dimensional problems. Using simulated data, we show that our approach could correctly capture temporal dependencies in firing rates and identify synchronous neurons. We also apply our model to spike train data obtained from prefrontal cortical areas.

---

65. A Bayesian Model of Polychronicity

Mira Guise, Alistair Knott, ■ubica Be■ušková
https://doi.org/10.1162/neco_a_00620 | pp. 2052-2073

A significant feature of spiking neural networks with varying connection delays, such as those in the brain, is the existence of strongly connected groups of neurons known as polychronous neural groups (PNGs). Polychronous groups are found in large numbers in these networks and are proposed by Izhikevich (2006a) to provide a neural basis for representation and memory. When exposed to a familiar stimulus, spiking neural networks produce consistencies in the spiking output data that are the hallmarks of PNG activation. Previous methods for studying the PNG activation response to stimuli have been limited by the template-based methods used to identify PNG activation. In this letter, we outline a new method that overcomes these difficulties by establishing for the first time a probabilistic interpretation of PNG activation. We then demonstrate the use of this method by investigating the claim that PNGs might provide the foundation of a representational system.

---

66. A Nonparametric Clustering Algorithm with a Quantile-Based Likelihood Estimator

Hideitsu Hino, Noboru Murata
https://doi.org/10.1162/neco_a_00628 | pp. 2074-2101

Clustering is a representative of unsupervised learning and one of the important approaches in exploratory data analysis. By its very nature, clustering without strong assumption on data distribution is desirable. Information-theoretic clustering is a class of clustering methods that optimize information-theoretic quantities such as entropy and mutual information. These quantities can be estimated in a nonparametric manner, and information-theoretic clustering algorithms are capable of capturing various intrinsic data structures. It is also possible to estimate information-theoretic quantities using a data set with sampling weight for each datum. Assuming the data set is sampled from a certain cluster and assigning different sampling weights depending on the clusters, the cluster-conditional information-theoretic quantities are estimated. In this letter, a simple iterative clustering algorithm is proposed based on a nonparametric estimator of the log likelihood for weighted data sets. The clustering algorithm is also derived from the principle of conditional entropy minimization with maximum entropy regularization. The proposed algorithm does not contain a tuning parameter. The algorithm is experimentally shown to be comparable to or outperform conventional nonparametric clustering methods.

---

67. Efficient Sensory Encoding and Bayesian Inference with Heterogeneous Neural Populations

Deep Ganguli, Eero P. Simoncelli
https://doi.org/10.1162/neco_a_00638 | pp. 2103-2134

The efficient coding hypothesis posits that sensory systems maximize information transmitted to the brain about the environment. We develop a precise and testable form of this hypothesis in the context of encoding a sensory variable with a population of noisy neurons, each characterized by a tuning curve. We parameterize the population with two continuous functions that control the density and amplitude of the tuning curves, assuming that the tuning widths vary inversely with the cell density. This parameterization

allows us to solve, in closed form, for the information-maximizing allocation of tuning curves as a function of the prior probability distribution of sensory variables. For the optimal population, the cell density is proportional to the prior, such that more cells with narrower tuning are allocated to encode higher-probability stimuli and that each cell transmits an equal portion of the stimulus probability mass. We also compute the stimulus discrimination capabilities of a perceptual system that relies on this neural representation and find that the best achievable discrimination thresholds are inversely proportional to the sensory prior. We examine how the prior information that is implicitly encoded in the tuning curves of the optimal population may be used for perceptual inference and derive a novel decoder, the Bayesian population vector, that closely approximates a Bayesian least-squares estimator that has explicit access to the prior. Finally, we generalize these results to sigmoidal tuning curves, correlated neural variability, and a broader class of objective functions. These results provide a principled embedding of sensory prior information in neural populations and yield predictions that are readily testable with environmental, physiological, and perceptual data.

---

68. Dimensionality of Object Representations in Monkey Inferotemporal Cortex

Sidney R. Lehky, Roozbeh Kiani, Hossein Esteky, Keiji Tanaka

https://doi.org/10.1162/neco_a_00648 | pp. 2135-2162

We have calculated the intrinsic dimensionality of visual object representations in anterior inferotemporal (AIT) cortex, based on responses of a large sample of cells stimulated with photographs of diverse objects. Because dimensionality was dependent on data set size, we determined asymptotic dimensionality as both the number of neurons and number of stimulus image approached infinity. Our final dimensionality estimate was 93 (SD: ± 11), indicating that there is basis set of approximately 100 independent features that characterize the dimensions of neural object space. We believe this is the first estimate of the dimensionality of neural visual representations based on single-cell neurophysiological data. The dimensionality of AIT object representations was much lower than the dimensionality of the stimuli. We suggest that there may be a gradual reduction in the dimensionality of object representations in neural populations going from retina to inferotemporal cortex as receptive fields become increasingly complex.

---

69. Hierarchical Control Using Networks Trained with Higher-Level Forward Models

Greg Wayne, L. F. Abbott

https://doi.org/10.1162/neco_a_00639 | pp. 2163-2193

We propose and develop a hierarchical approach to network control of complex tasks. In this approach, a low-level controller directs the activity of a "plant," the system that performs the task. However, the low-level controller may be able to solve only fairly simple problems involving the plant. To accomplish more complex tasks, we introduce a higher-level controller that controls the lower-level controller. We use this system to direct an articulated truck to a specified location through an environment filled with static or moving obstacles. The final system consists of networks that have memorized associations between the sensory data they receive and the commands they issue. These networks are trained on a set of optimal associations generated by minimizing cost functions. Cost function minimization requires predicting the

consequences of sequences of commands, which is achieved by constructing forward models, including a model of the lower-level controller. The forward models and cost minimization are used only during training, allowing the trained networks to respond rapidly. In general, the hierarchical approach can be extended to larger numbers of levels, dividing complex tasks into more manageable subtasks. The optimization procedure and the construction of the forward models and controllers can be performed in similar ways at each level of the hierarchy, which allows the system to be modified to perform other tasks or to be extended for more complex tasks without retraining lower-levels.

---

70. Spine Head Calcium as a Measure of Summed Postsynaptic Activity for Driving Synaptic Plasticity

Bruce Graham, Aušra Saudargien■, Stuart Cobb

https://doi.org/10.1162/neco_a_00640 | pp. 2194-2222

We use a computational model of a hippocampal CA1 pyramidal cell to demonstrate that spine head calcium provides an instantaneous readout at each synapse of the postsynaptic weighted sum of all presynaptic activity impinging on the cell. The form of the readout is equivalent to the functions of weighted, summed inputs used in neural network learning rules. Within a dendritic layer, peak spine head calcium levels are either a linear or sigmoidal function of the number of coactive synapses, with nonlinearity depending on the ability of voltage spread in the dendrites to reach calcium spike threshold. This is strongly controlled by the potassium A-type current, with calcium spikes and the consequent sigmoidal increase in peak spine head calcium present only when the A-channel density is low. Other membrane characteristics influence the gain of the relationship between peak calcium and the number of active synapses. In particular, increasing spine neck resistance increases the gain due to increased voltage responses to synaptic input in spine heads. Colocation of stimulated synapses on a single dendritic branch also increases the gain of the response. Input pathways cooperate: CA3 inputs to the proximal apical dendrites can strongly amplify peak calcium levels due to weak EC input to the distal dendrites, but not so strongly vice versa. CA3 inputs to the basal dendrites can boost calcium levels in the proximal apical dendrites, but the relative electrical compactness of the basal dendrites results in the reverse effect being less significant. These results give pointers as to how to better describe the contributions of pre- and postsynaptic activity in the learning "rules" that apply in these cells. The calcium signal is closer in form to the activity measures used in traditional neural network learning rules than to the spike times used in spike-timing-dependent plasticity.

---

71. Design of Charge-Balanced Time-Optimal Stimuli for Spiking Neuron Oscillators

Isuru Dasanayake, Jr-Shin Li

https://doi.org/10.1162/neco_a_00643 | pp. 2223-2246

In this letter, we investigate the fundamental limits on how the interspike time of a neuron oscillator can be perturbed by the application of a bounded external control input (a current stimulus) with zero net electric charge accumulation. We use phase models to study the dynamics of neurons and derive charge-balanced controls that achieve the minimum and maximum interspike times for a given bound on the control amplitude. Our derivation is valid for any arbitrary shape of the phase response curve and for any value of the given control amplitude bound. In addition, we characterize the change in the structures of the

charge-balanced time-optimal controls with the allowable control amplitude. We demonstrate the applicability of the derived optimal control laws by applying them to mathematically ideal and experimentally observed neuron phase models, including the widely studied Hodgkin-Huxley phase model, and by verifying them with the corresponding original full state-space models. This work addresses a fundamental problem in the field of neural control and provides a theoretical investigation to the optimal control of oscillatory systems.

---

72. Influence of External Inputs and Asymmetry of Connections on Information-Geometric Measures Involving Up to Ten Neuronal Interactions

Yimin Nie, Jean■Marc Fellous, Masami Tatsuno

https://doi.org/10.1162/neco_a_00633 | pp. 2247-2293

The investigation of neural interactions is crucial for understanding information processing in the brain. Recently an analysis method based on information geometry (IG) has gained increased attention, and the property of the pairwise IG measure has been studied extensively in relation to the two-neuron interaction. However, little is known about the property of IG measures involving more neuronal interactions. In this study, we systematically investigated the influence of external inputs and the asymmetry of connections on the IG measures in cases ranging from 1-neuron to 10-neuron interactions. First, the analytical relationship between the IG measures and external inputs was derived for a network of 10 neurons with uniform connections. Our results confirmed that the single and pairwise IG measures were good estimators of the mean background input and of the sum of the connection weights, respectively. For the IG measures involving 3 to 10 neuronal interactions, we found that the influence of external inputs was highly nonlinear. Second, by computer simulation, we extended our analytical results to asymmetric connections. For a network of 10 neurons, the simulation showed that the behavior of the IG measures in relation to external inputs was similar to the analytical solution obtained for a uniformly connected network. When the network size was increased to 1000 neurons, the influence of external inputs almost disappeared. This result suggests that all IG measures from 1-neuron to 10-neuron interactions are robust against the influence of external inputs. In addition, we investigated how the strength of asymmetry influenced the IG measures. Computer simulation of a 1000-neuron network showed that all the IG measures were robust against the modulation of the asymmetry of connections. Our results provide further support for an information-geometric approach and will provide useful insights when these IG measures are applied to real experimental spike data.

---

73. Fragility in Dynamic Networks: Application to Neural Networks in the Epileptic Cortex

Duluxan Sritharan, Sridevi V. Sarma

https://doi.org/10.1162/neco_a_00644 | pp. 2294-2327

Epilepsy is a network phenomenon characterized by atypical activity at the neuronal and population levels during seizures, including tonic spiking, increased heterogeneity in spiking rates, and synchronization. The etiology of epilepsy is unclear, but a common theme among proposed mechanisms is that structural connectivity between neurons is altered. It is hypothesized that epilepsy arises not from random changes in connectivity, but from specific structural changes to the most fragile nodes or neurons in the network. In this

letter, the minimum energy perturbation on functional connectivity required to destabilize linear networks is derived. Perturbation results are then applied to a probabilistic nonlinear neural network model that operates at a stable fixed point. That is, if a small stimulus is applied to the network, the activation probabilities of each neuron respond transiently but eventually recover to their baseline values. When the perturbed network is destabilized, the activation probabilities shift to larger or smaller values or oscillate when a small stimulus is applied. Finally, the structural modifications to the neural network that achieve the functional perturbation are derived. Simulations of the unperturbed and perturbed networks qualitatively reflect neuronal activity observed in epilepsy patients, suggesting that the changes in network dynamics due to destabilizing perturbations, including the emergence of an unstable manifold or a stable limit cycle, may be indicative of neuronal or population dynamics during seizure. That is, the epileptic cortex is always on the brink of instability and minute changes in the synaptic weights associated with the most fragile node can suddenly destabilize the network to cause seizures. Finally, the theory developed here and its interpretation of epileptic networks enables the design of a straightforward feedback controller that first detects when the network has destabilized and then applies linear state feedback control to steer the network back to its stable state.

---

74. On Antiperiodic Solutions for Cohen-Grossberg Shunting Inhibitory Neural Networks with Time-Varying Delays and Impulses

Changjin Xu, Qiming Zhang

https://doi.org/10.1162/neco_a_00642 | pp. 2328-2349

In this letter, a class of Cohen-Grossberg shunting inhibitory neural networks with time-varying delays and impulses is investigated. Sufficient conditions for the existence and exponential stability of antiperiodic solutions of such a class of neural networks are established. Our results are new and complementary to previously known results. An example is given to illustrate the feasibility and effectiveness of our main results.

---

75. Learning Rates ofl^qCoefficient Regularization Learning with Gaussian Kernel

Shaobo Lin, Jinshan Zeng, Jian Fang, Zongben Xu

https://doi.org/10.1162/neco_a_00641 | pp. 2350-2378

Regularization is a well-recognized powerful strategy to improve the performance of a learning machine and l(q) regularization schemes with $0 < q < \infty$ are central in use. It is known that different q leads to different properties of the deduced estimators, say, l(2) regularization leads to a smooth estimator, while l(1) regularization leads to a sparse estimator. Then how the generalization capability of l(q) regularization learning varies with q is worthy of investigation. In this letter, we study this problem in the framework of statistical learning theory. Our main results show that implementing l(q) coefficient regularization schemes in the sample-dependent hypothesis space associated with a gaussian kernel can attain the same almost optimal learning rates for all $0 < q < \infty$. That is, the upper and lower bounds of learning rates for l(q) regularization learning are asymptotically identical for all $0 < q < \infty$. Our finding tentatively reveals that in some modeling contexts, the choice of q might not have a strong impact on the generalization capability. From this perspective, q can be arbitrarily specified, or specified merely by other nongeneralization criteria

like smoothness, computational complexity or sparsity.

---

76. High-Dimensional Cluster Analysis with the Masked EM Algorithm

Shabnam Kadir, Dan F. M. Goodman, Kenneth D. Harris

https://doi.org/10.1162/neco_a_00661 | pp. 2379-2394

Cluster analysis faces two problems in high dimensions: the "curse of dimensionality" that can lead to overfitting and poor generalization performance and the sheer time taken for conventional algorithms to process large amounts of high-dimensional data. We describe a solution to these problems, designed for the application of spike sorting for next-generation, high-channel-count neural probes. In this problem, only a small subset of features provides information about the cluster membership of any one data vector, but this informative feature subset is not the same for all data points, rendering classical feature selection ineffective. We introduce a "masked EM" algorithm that allows accurate and time-efficient clustering of up to millions of points in thousands of dimensions. We demonstrate its applicability to synthetic data and to real-world high-channel-count spike sorting data.

---

77. Changes of Firing Rate Induced by Changes of Phase Response Curve in Bifurcation Transitions

Yasuomi D. Sato, Kazuyuki Aihara

https://doi.org/10.1162/neco_a_00653 | pp. 2395-2418

We study dynamical mechanisms responsible for changes of the firing rate during four different bifurcation transitions in the two-dimensional Hindmarsh-Rose (2DHR) neuron model: the saddle node on an invariant circle (SNIC) bifurcation to the supercritical Andronov-Hopf (AH) one, the SNIC bifurcation to the saddle-separatrix loop (SSL) one, the AH bifurcation to the subcritical AH (SAH) one, and the SSL bifurcation to the AH one. For this purpose, we study slopes of the firing rate curve with respect to not only an external input current but also temperature that can be interpreted as a timescale in the 2DHR neuron model. These slopes are mathematically formulated with phase response curves (PRCs), expanding the firing rate with perturbations of the temperature and external input current on the one-dimensional space of the phase [Formula: see text] in the 2DHR oscillator. By analyzing the two different slopes of the firing rate curve with respect to the temperature and external input current, we find that during changes of the firing rate in all of the bifurcation transitions, the calculated slope with respect to the temperature also changes. This is largely dependent on changes in the PRC size that is also related to the slope with respect to the external input current. Furthermore, we find phase transition-like switches of the firing rate with a possible increase of the temperature during the SSL-to-AH bifurcation transition.

---

78. Approximate Emergent Synchrony in Spatially Coupled Spiking Neurons with Discrete Interaction

Hans Supèr, August Romeo

https://doi.org/10.1162/neco_a_00658 | pp. 2419-2440

Models for perceptual grouping and contour integration are presented. Connection weights depend on distances and angle differences, while neurons evolve following a spiking dynamics (Izhikevich's model in most of the considered cases). Although the studied synapses depend on discrete three-valued functions, simulations display the emergence of approximate synchrony, making these cognitive tasks possible. Noise effects are examined, and the possibility of achieving similar results with a different neuron model is discussed.

---

79. Facilitation of Neuronal Responses by Intrinsic Default Mode Network Activity

Hiroakira Matsui, Meihong Zheng, Osamu Hoshino

https://doi.org/10.1162/neco_a_00660 | pp. 2441-2464

Default mode network (DMN) shows intrinsic, high-level activity at rest. We tested a hypothesis proposed for its role in sensory information processing: Intrinsic DMN activity facilitates neural responses to sensory input. A neural network model, consisting of a sensory network (Nsen) and a DMN, was simulated. The Nsen contained cell assemblies. Each cell assembly comprised principal cells, GABAergic interneurons (Ia, Ib), and glial cells. We let the Nsen carry out a perceptual task: detection of sensory stimuli. During DMN activation, glial cells were hyperpolarized by Ia-to-glia circuitry, by which glial membrane transporters imported GABA molecules from the extracellular space and decreased ambient GABA concentration. Acting on extrasynaptic GABA receptors, the decrease in ambient GABA concentration reduced inhibitory current in a tonic manner. This depolarized principal cells below their firing threshold during the ongoing spontaneous time period and accelerated their reaction speed to a sensory stimulus. During the stimulus presentation period, the Nsen inhibited the DMN and caused DMN deactivation. The DMN deactivation made Nsen Ia cells cease firing, thereby stopping the glial membrane hyperpolarization, quitting the GABA import, returning to the basal ambient GABA level, and thus enhancing global inhibition. Notably, the stimulus-relevant P cell firing could be maintained when GABAergic gliotransmission via Ia-glia signaling worked, decreasing ambient GABA concentration around the stimulus-relevant P cells. This enabled the Nsen to reliably detect the stimulus. We suggest that intrinsic default model network activity may accelerate the reaction speed of the sensory network by modulating its ongoing-spontaneous activity in a subthreshold manner. Ambient GABA contributes to achieve an optimal ongoing spontaneous subthreshold neuronal state, in which GABAergic gliotransmission triggered by the intrinsic default model network activity may play an important role.

---

80. A Hierarchical Adaptive Approach to Optimal Experimental Design

Woojae Kim, Mark A. Pitt, Zhong■Lin Lu, Mark Steyvers, Jay I. Myung

https://doi.org/10.1162/neco_a_00654 | pp. 2465-2492

Experimentation is at the core of research in the behavioral and neural sciences, yet observations can be expensive and time-consuming to acquire (e.g., MRI scans, responses from infant participants). A major interest of researchers is designing experiments that lead to maximal accumulation of information about the phenomenon under study with the fewest possible number of observations. In addressing this challenge, statisticians have developed adaptive design optimization methods. This letter introduces a hierarchical Bayes extension of adaptive design optimization that provides a judicious way to exploit two complementary

schemes of inference (with past and future data) to achieve even greater accuracy and efficiency in information gain. We demonstrate the method in a simulation experiment in the field of visual perception.

---

81. Noise Facilitation in Associative Memories of Exponential Capacity

Amin Karbasi, Amir Hesam Salavati, Amin Shokrollahi, Lav R. Varshney

https://doi.org/10.1162/neco_a_00655 | pp. 2493-2526

Recent advances in associative memory design through structured pattern sets and graph-based inference algorithms have allowed reliable learning and recall of an exponential number of patterns that satisfy certain subspace constraints. Although these designs correct external errors in recall, they assume neurons that compute noiselessly, in contrast to the highly variable neurons in brain regions thought to operate associatively, such as hippocampus and olfactory cortex. Here we consider associative memories with boundedly noisy internal computations and analytically characterize performance. As long as the internal noise level is below a specified threshold, the error probability in the recall phase can be made exceedingly small. More surprising, we show that internal noise improves the performance of the recall phase while the pattern retrieval capacity remains intact: the number of stored patterns does not reduce with noise (up to a threshold). Computational experiments lend additional support to our theoretical analysis. This work suggests a functional benefit to noisy neurons in biological neuronal networks.

---

82. A No-Go Theorem for One-Layer Feedforward Networks

Chad Giusti, Vladimir Itskov

https://doi.org/10.1162/neco_a_00657 | pp. 2527-2540

It is often hypothesized that a crucial role for recurrent connections in the brain is to constrain the set of possible response patterns, thereby shaping the neural code. This implies the existence of neural codes that cannot arise solely from feedforward processing. We set out to find such codes in the context of one-layer feedforward networks and identified a large class of combinatorial codes that indeed cannot be shaped by the feedforward architecture alone. However, these codes are difficult to distinguish from codes that share the same sets of maximal activity patterns in the presence of subtractive noise. When we coarsened the notion of combinatorial neural code to keep track of only maximal patterns, we found the surprising result that all such codes can in fact be realized by one-layer feedforward networks. This suggests that recurrent or many-layer feedforward architectures are not necessary for shaping the (coarse) combinatorial features of neural codes. In particular, it is not possible to infer a computational role for recurrent connections from the combinatorics of neural response patterns alone. Our proofs use mathematical tools from classical combinatorial topology, such as the nerve lemma and the existence of an inverse nerve. An unexpected corollary of our main result is that any prescribed (finite) homotopy type can be realized by a subset of the form [Formula: see text], where P is a polyhedron.

---

83. Extended Robust Support Vector Machine Based on Financial Risk Minimization

Akiko Takeda, Shuhei Fujiwara, Takafumi Kanamori

https://doi.org/10.1162/neco_a_00647 | pp. 2541-2569

Financial risk measures have been used recently in machine learning. For example, ν-support vector machine ν-SVM) minimizes the conditional value at risk (CVaR) of margin distribution. The measure is popular in finance because of the subadditivity property, but it is very sensitive to a few outliers in the tail of the distribution. We propose a new classification method, extended robust SVM (ER-SVM), which minimizes an intermediate risk measure between the CVaR and value at risk (VaR) by expecting that the resulting model becomes less sensitive than ν-SVM to outliers. We can regard ER-SVM as an extension of robust SVM, which uses a truncated hinge loss. Numerical experiments imply the ER-SVM's possibility of achieving a better prediction performance with proper parameter setting.

---

84. Nonparametric Estimation of Küllback-Leibler Divergence

Zhiyi Zhang, Michael Grabchak

https://doi.org/10.1162/neco_a_00646 | pp. 2570-2593

In this letter, we introduce an estimator of Küllback-Leibler divergence based on two independent samples. We show that on any finite alphabet, this estimator has an exponentially decaying bias and that it is consistent and asymptotically normal. To explain the importance of this estimator, we provide a thorough analysis of the more standard plug-in estimator. We show that it is consistent and asymptotically normal, but with an infinite bias. Moreover, if we modify the plug-in estimator to remove the rare events that cause the bias to become infinite, the bias still decays at a rate no faster than O(1/n). Further, we extend our results to estimating the symmetrized Küllback-Leibler divergence. We conclude by providing simulation results, which show that the asymptotic properties of these estimators hold even for relatively small sample sizes.

---

85. Coenzyme Q10 Benefits Symptoms in Gulf War Veterans: Results of a Randomized Double-Blind Study

Beatrice A. Golomb, Matthew Allison, Sabrina Koperski, Hayley J. Koslik, Sridevi Devaraj, Janis B. Ritchie

https://doi.org/10.1162/neco_a_00659 | pp. 2594-2651

We sought to assess whether coenzyme Q10 (CoQ10) benefits the chronic multisymptom problems that affect one-quarter to one-third of 1990-1 Gulf War veterans, using a randomized, double-blind, placebo-controlled study. Participants were 46 veterans meeting Kansas and Centers for Disease Control criteria for Gulf War illness. Intervention was PharmaNord (Denmark) CoQ10 100 mg per day (Q100), 300 mg per day (Q300), or an identical-appearing placebo for 3.5 ± 0.5 months. General self-rated health (GSRH), the primary outcome, differed across randomization arms at baseline, and sex significantly predicted GSRH change, compelling adjustment for baseline GSRH and prompting sex-stratified analysis. GSRH showed no significant benefit in the combined-sex sample. Among males (85% of participants), Q100 significantly benefited GSRH versus placebo and versus Q300, providing emphasis on Q100. Physical function (summary performance score, SPS) improved on Q100 versus placebo. A rise in CoQ10 approached significance as a predictor of improvement in GSRH and significantly predicted SPS improvement. Among 20 symptoms each present in half or more of the enrolled veterans, direction-of-difference on Q100 versus placebo was favorable for all except sleep problems; sign test 19:1, p=0.00004) with several symptoms individually significant. Significance for these symptoms despite the

small sample underscores large effect sizes, and an apparent relation of key outcomes to CoQ10 change increases prospects for causality. In conclusion, Q100 conferred benefit to physical function and symptoms in veterans with Gulf War illness. Examination in a larger sample is warranted, and findings from this study can inform the conduct of a larger trial.

---

86. Visual Motion Priors Differ for Infants and Mothers

Florian Raudies, Rick O. Gilmore

https://doi.org/10.1162/neco_a_00645 | pp. 2652-2668

Visual motion direction ambiguities due to edge-aperture interaction might be resolved by speed priors, but scant empirical data support this hypothesis. We measured optic flow and gaze positions of walking mothers and the infants they carried. Empirically derived motion priors for infants are vertically elongated and shifted upward relative to mothers. Skewed normal distributions fitted to estimated retinal speeds peak at values above 20°/sec.

---

87. Risk-Aware Control

Terence D. Sanger

https://doi.org/10.1162/neco_a_00662 | pp. 2669-2691

Human movement differs from robot control because of its flexibility in unknown environments, robustness to perturbation, and tolerance of unknown parameters and unpredictable variability. We propose a new theory, risk-aware control, in which movement is governed by estimates of risk based on uncertainty about the current state and knowledge of the cost of errors. We demonstrate the existence of a feedback control law that implements risk-aware control and show that this control law can be directly implemented by populations of spiking neurons. Simulated examples of risk-aware control for time-varying cost functions as well as learning of unknown dynamics in a stochastic risky environment are provided.

---

88. A Neural Framework for Organization and Flexible Utilization of Episodic Memory in Cumulatively Learning Baby Humanoids

Vishwanathan Mohan, Giulio Sandini, Pietro Morasso

https://doi.org/10.1162/neco_a_00664 | pp. 2692-2734

Cumulatively developing robots offer a unique opportunity to reenact the constant interplay between neural mechanisms related to learning, memory, prospection, and abstraction from the perspective of an integrated system that acts, learns, remembers, reasons, and makes mistakes. Situated within such interplay lie some of the computationally elusive and fundamental aspects of cognitive behavior: the ability to recall and flexibly exploit diverse experiences of one's past in the context of the present to realize goals, simulate the future, and keep learning further. This article is an adventurous exploration in this direction using a simple engaging scenario of how the humanoid iCub learns to construct the tallest possible stack given an arbitrary set of objects to play with. The learning takes place cumulatively, with the robot interacting with different objects (some previously experienced, some novel) in an open-ended fashion. Since the solution itself depends on what objects are available in the "now," multiple episodes of past experiences have to be remembered and

creatively integrated in the context of the present to be successful. Starting from zero, where the robot knows nothing, we explore the computational basis of organization episodic memory in a cumulatively learning humanoid and address (1) how relevant past experiences can be reconstructed based on the present context, (2) how multiple stored episodic memories compete to survive in the neural space and not be forgotten, (3) how remembered past experiences can be combined with explorative actions to learn something new, and (4) how multiple remembered experiences can be recombined to generate novel behaviors (without exploration). Through the resulting behaviors of the robot as it builds, breaks, learns, and remembers, we emphasize that mechanisms of episodic memory are fundamental design features necessary to enable the survival of autonomous robots in a real world where neither everything can be known nor can everything be experienced.

---

89. Computing with a Canonical Neural Circuits Model with Pool Normalization and Modulating Feedback

Tobias Brosch, Heiko Neumann

https://doi.org/10.1162/neco_a_00675 | pp. 2735-2789

Evidence suggests that the brain uses an operational set of canonical computations like normalization, input filtering, and response gain enhancement via reentrant feedback. Here, we propose a three-stage columnar architecture of cascaded model neurons to describe a core circuit combining signal pathways of feedforward and feedback processing and the inhibitory pooling of neurons to normalize the activity. We present an analytical investigation of such a circuit by first reducing its detail through the lumping of initial feedforward response filtering and reentrant modulating signal amplification. The resulting excitatory-inhibitory pair of neurons is analyzed in a 2D phase-space. The inhibitory pool activation is treated as a separate mechanism exhibiting different effects. We analyze subtractive as well as divisive (shunting) interaction to implement center-surround mechanisms that include normalization effects in the characteristics of real neurons. Different variants of a core model architecture are derived and analyzed--in particular, individual excitatory neurons (without pool inhibition), the interaction with an inhibitory subtractive or divisive (i.e., shunting) pool, and the dynamics of recurrent self-excitation combined with divisive inhibition. The stability and existence properties of these model instances are characterized, which serve as guidelines to adjust these properties through proper model parameterization. The significance of the derived results is demonstrated by theoretical predictions of response behaviors in the case of multiple interacting hypercolumns in a single and in multiple feature dimensions. In numerical simulations, we confirm these predictions and provide some explanations for different neural computational properties. Among those, we consider orientation contrast-dependent response behavior, different forms of attentional modulation, contrast element grouping, and the dynamic adaptation of the silent surround in extraclassical receptive field configurations, using only slight variations of the same core reference model.

---

90. On Quadrature Methods for Refractory Point Process Likelihoods

Gonzalo E. Mena, Liam Paninski

https://doi.org/10.1162/neco_a_00676 | pp. 2790-2797

Parametric models of the conditional intensity of a point process (e.g., generalized linear models) are popular in statistical neuroscience, as they allow us to characterize the variability in neural responses in

terms of stimuli and spiking history. Parameter estimation in these models relies heavily on accurate evaluations of the log likelihood and its derivatives. Classical approaches use a discretized time version of the spiking process, and recent work has exploited the existence of a refractory period (during which the conditional intensity is zero following a spike) to obtain more accurate estimates of the likelihood. In this brief letter, we demonstrate that this method can be improved significantly by applying classical quadrature methods directly to the resulting continuous-time integral.

---

91. Dynamics and Reliability of Bistable Neurons Driven with Time-Dependent Stimuli

Soledad Gonzalo Cogno, Susanne Schreiber, Inés Samengo
https://doi.org/10.1162/neco_a_00671 | pp. 2798-2826

The reliability of a spiking neuron depends on the frequency content of the driving input signal. Previous studies have shown that well above threshold, regularly firing neurons generate reliable responses when the input signal resonates with the firing frequency of the cell. Instead, well below threshold, reliable responses are obtained when the input frequency resonates with the subthreshold oscillations of the neuron. Previous theories, however, provide no clear prediction for the input frequency giving rise to maximally reliable spiking at threshold, which is probably the most relevant firing regime in mammalian cortex under physiological conditions. In particular, when the firing onset is governed by a subcritical Hopf bifurcation, the frequency of subthreshold oscillations often differs from the firing rate at threshold. The predictions of previous studies, hence, cannot be smoothly merged at threshold. Here we explore the behavior of reliability in bistable neurons near threshold using three types of driving stimuli: constant, periodic, and stochastic. We find that the two natural frequencies of the system, associated with the two coexisting attractors, provide a rich variety of possible locking modes with the external signal. Reliability is determined by the sensitivity to noise of each locking mode and by the transition probabilities between modes. Noise increases the amount of spike time jitter, and minimal jitter is obtained for input frequencies coinciding with the suprathreshold firing rate of the cell. In addition, noise may either enhance or inhibit transitions between the two attractors, depending on the input frequency. The dual role played by noise in bistable systems implies that reliability is determined by a delicate balance between spike time jitter and the rate of transitions between attractors.

---

92. A DC Programming Approach for Finding Communities in Networks

Hoai An Le Thi, Manh Cuong Nguyen, Tao Pham Dinh
https://doi.org/10.1162/neco_a_00673 | pp. 2827-2854

Automatic discovery of community structures in complex networks is a fundamental task in many disciplines, including physics, biology, and the social sciences. The most used criterion for characterizing the existence of a community structure in a network is modularity, a quantitative measure proposed by Newman and Girvan (2004). The discovery community can be formulated as the so-called modularity maximization problem that consists of finding a partition of nodes of a network with the highest modularity. In this letter, we propose a fast and scalable algorithm called DCAM, based on DC (difference of convex function) programming and DCA (DC algorithms), an innovative approach in nonconvex programming framework for solving the modularity maximization problem. The special structure of the problem considered here has been well exploited to get an inexpensive DCA scheme that requires only a matrix-vector product at each

iteration. Starting with a very large number of communities, DCAM furnishes, as output results, an optimal partition together with the optimal number of communities [Formula: see text]; that is, the number of communities is discovered automatically during DCAM's iterations. Numerical experiments are performed on a variety of real-world network data sets with up to 4,194,304 nodes and 30,359,198 edges. The comparative results with height reference algorithms show that the proposed approach outperforms them not only on quality and rapidity but also on scalability. Moreover, it realizes a very good trade-off between the quality of solutions and the run time.

---

93. Alternating Proximal Regularized Dictionary Learning

Saverio Salzo, Salvatore Masecchia, Alessandro Verri, Annalisa Barla

https://doi.org/10.1162/neco_a_00672 | pp. 2855-2895

We present an algorithm for dictionary learning that is based on the alternating proximal algorithm studied by Attouch, Bolte, Redont, and Soubeyran (2010), coupled with a reliable and efficient dual algorithm for computation of the related proximity operators. This algorithm is suitable for a general dictionary learning model composed of a Bregman-type data fit term that accounts for the goodness of the representation and several convex penalization terms on the coefficients and atoms, explaining the prior knowledge at hand. As Attouch et al. recently proved, an alternating proximal scheme ensures better convergence properties than the simpler alternating minimization. We take care of the issue of inexactness in the computation of the involved proximity operators, giving a sound stopping criterion for the dual inner algorithm, which keeps under control the related errors, unavoidable for such a complex penalty terms, providing ultimately an overall effective procedure. Thanks to the generality of the proposed framework, we give an application in the context of genome-wide data understanding, revising the model proposed by Nowak, Hastie, Pollack, and Tibshirani (2011). The aim is to extract latent features (atoms) and perform segmentation on array-based comparative genomic hybridization (aCGH) data. We improve several important aspects that increase the quality and interpretability of the results. We show the effectiveness of the proposed model with two experiments on synthetic data, which highlight the enhancements over the original model.

---

94. U-Processes and Preference Learning

Hong Li, Chuanbao Ren, Luoqing Li

https://doi.org/10.1162/neco_a_00674 | pp. 2896-2924

Preference learning has caused great attention in machining learning. In this letter we propose a learning framework for pairwise loss based on empirical risk minimization of U-processes via Rademacher complexity. We first establish a uniform version of Bernstein inequality of U-processes of degree 2 via the entropy methods. Then we estimate the bound of the excess risk by using the Bernstein inequality and peeling skills. Finally, we apply the excess risk bound to the pairwise preference and derive the convergence rates of pairwise preference learning algorithms with squared loss and indicator loss by using the empirical risk minimization with respect to U-processes.

---

95. Spiking Neural P Systems with a Generalized Use of Rules

Xingyi Zhang, Bangju Wang, Linqiang Pan
https://doi.org/10.1162/neco_a_00665 | pp. 2925-2943

Spiking neural P systems (SN P systems) are a class of distributed parallel computing devices inspired by spiking neurons, where the spiking rules are usually used in a sequential way (an applicable rule is applied one time at a step) or an exhaustive way (an applicable rule is applied as many times as possible at a step). In this letter, we consider a generalized way of using spiking rules by "combining" the sequential way and the exhaustive way: if a rule is used at some step, then at that step, it can be applied any possible number of times, nondeterministically chosen. The computational power of SN P systems with a generalized use of rules is investigated. Specifically, we prove that SN P systems with a generalized use of rules consisting of one neuron can characterize finite sets of numbers. If the systems consist of two neurons, then the computational power of such systems can be greatly improved, but not beyond generating semilinear sets of numbers. SN P systems with a generalized use of rules consisting of three neurons are proved to generate at least a non-semilinear set of numbers. In the case of allowing enough neurons, SN P systems with a generalized use of rules are computationally complete. These results show that the number of neurons is crucial for SN P systems with a generalized use of rules to achieve a desired computational power.

---

96. Sequence Memory Based on Coherent Spin-Interaction Neural Networks

Min Xia, Wai Keung Wong, Zhijie Wang
https://doi.org/10.1162/neco_a_00663 | pp. 2944-2961

Sequence information processing, for instance, the sequence memory, plays an important role on many functions of brain. In the workings of the human brain, the steady-state period is alterable. However, in the existing sequence memory models using heteroassociations, the steady-state period cannot be changed in the sequence recall. In this work, a novel neural network model for sequence memory with controllable steady-state period based on coherent spininteraction is proposed. In the proposed model, neurons fire collectively in a phase-coherent manner, which lets a neuron group respond differently to different patterns and also lets different neuron groups respond differently to one pattern. The simulation results demonstrating the performance of the sequence memory are presented. By introducing a new coherent spin-interaction sequence memory model, the steady-state period can be controlled by dimension parameters and the overlap between the input pattern and the stored patterns. The sequence storage capacity is enlarged by coherent spin interaction compared with the existing sequence memory models. Furthermore, the sequence storage capacity has an exponential relationship to the dimension of the neural network.